Orthopolity: a general law of resource allocation

Resource capacity, complete hierarchies, and effective-resource inference

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Abstract

Orthopolity expresses a general resource-allocation principle: Nature tends to distribute resources equally among concentrations of those resources. We establish its mathematical foundation in finite resource capacity and complete hierarchies. A resource budget bounds the number of units above any resource threshold; at every complete partition level, total resource is identical and count is inversely proportional to mean resource per unit. A coverage identity extends this result to incomplete and overlapping descriptions. These exact consequences of additive accounting anchor a general natural-law formulation whose expression depends on physical constraints and the comparison measure. Their extension to equal allocation among coexisting resource classes requires physical conditions. Effective resources may combine constituents nonlinearly; we derive a restriction on combinations that remain invariant under subdivision. Forward allocation and inverse inference then connect resource requirements to abundance. A shared product-resource family supplies identification conditions and predictions across environments, demonstrated synthetically. Class exchange proves equalization, constrained equilibrium and recovery. Electromagnetic transport and thermal radiation supply established physical realizations. Growth and division connect a population's mass allocation to lineage sampling; an independently calibrated bacterial comparison predicts mean length closely but overpredicts variability. Retained studies include conditional budget-closure successes and failed response forecasts. A frozen linguistic application predicts withheld genres better with combined letter/phoneme cost than with letter-only models, while lexical frequencies remain more predictive. The contribution connects an exact capacity and hierarchy principle to a testable allocation framework. Its component mathematics retains its established foundations; the broader physical scope of orthopolity remains an empirical claim.

1. Introduction

Orthopolity names a general principle connecting the resource held by a unit to the number of such units that a system can support. Its natural-law formulation is:

Nature tends to distribute resources equally among concentrations of those resources.

The elementary intuition is geometric. A box of volume V accommodates V/v smaller boxes of volume v when they tile it exactly. Halving the side of three-dimensional cubes multiplies their number by eight while leaving their combined volume unchanged. The same relation holds at every level of a complete nested subdivision. More generally, for a fixed additive budget,

increasing the requirement per unit lowers the budget's upper bound on count. This is the mathematical starting point, before a stochastic model or fitted distribution.

This paper establishes the resource-capacity bound and the complete-hierarchy equality under explicit assumptions, then develops **orthopolity as a general natural law** through mechanisms, known physical realizations, empirical comparisons and inverse inference. The elementary mathematics is established accounting; the generalizing claim concerns its organizing role across natural systems. A capacity bound, an equality across complete descriptions of the same system, and an abundance distribution among coexisting classes are related statements with different conditions. Section 2.1 connects them precisely. The broader claim of a physical allocation tendency remains subject to those conditions and to evidence about its scope. A simple mathematical foundation can have far-reaching physical consequences without making every extension an automatic theorem.

The term follows Fabbri's 2024 essay and an earlier manuscript by Fabbri and Oliveira Jr. (2017). The earlier account assumes uniform resource allocation across component-wealth values; the later essay proposes a broader natural and cosmological principle. Here we develop a common formulation while keeping each physical comparison explicit.

Real concentrations depend on several resources together. Algae develop within available space, light, water and multiple nutrients over time; cities depend on land, people, buildings, transport, energy, money, food and water. These quantities interact and may substitute for, complement, or limit one another. The operative resource may therefore be a nonlinear effective combination. Deriving it from individual inputs can be difficult. Orthopolity also motivates the reverse direction: use an observed allocation to infer properties of the effective resource, then test the resulting resource model through independent measurements or predictions in other conditions. Forward prediction and inverse inference are complementary uses of the law, developed in Sections 2.2–2.3.

The proposed tendency must be distinguished from the resulting distribution. A standing tree experiences gravity while its structure and anchorage support it. An apple thrown upward initially rises while gravity acts; catching it or lodging it against a ceiling introduces an interaction that can prevent its return to the ground. These outcomes reflect gravity together with other interactions and initial conditions. The same logical distinction guides orthopolity: its visible signature can be altered or obscured by the construction of the system and its medium. Human age and height distributions motivate analyses of such constraints. Their lack of an apparent neutral signature would not, by itself, establish a general absence of evidence for orthopolity in Nature. Section 2.7 states how this distinction enters the research programme.

The ecological precedent is substantial. Sheldon, Prakash and Sutcliffe (1972) studied ocean particle size distributions; Gaedke (1993) explicitly related equal biomass in logarithmic classes to normalized biomass spectra; Hatton et al. (2021) reconstructed the ocean spectrum from bacteria to whales. Arranz et al. (2022) studied variation among hundreds of lake fish communities and its ecological correlates. Variation among individual systems is therefore already a research topic. Energetic equivalence (Damuth, 1981) is related, but its species-population sampling unit differs from community abundance per size interval. Species richness across sizes prevents automatic interchange of those statements.

Power laws arise under different generative mechanisms, so an observed exponent supplies an inverse constraint rather than a unique causal explanation (Newman, 2005; Clauset, Shalizi and Newman, 2009). Our formulation specifies what a distribution can reveal about effective resources and what additional information resolves competing interpretations. The class-exchange model demonstrates an equalizing contribution that persists under a constraint. Electromagnetic transport and thermal radiation connect the proposed law to known physical equations, with different physically defined comparison measures. The empirical cases establish the successes and

limits of specific resource and response models. The accounting identity, reversible transport, classical equipartition and quantum radiation law are established results; their organization into a general allocation proposal and its forward and inverse programme is the contribution developed here. Inverse optimization already estimates objectives from observed decisions (Keshavarz, Wang and Boyd, 2011), and inverse statistical mechanics estimates interactions from ensemble properties (Habeck, 2014). Our inverse contribution concerns the resource interpretation and the consequences of sharing an allocation model across environments. A comparison with these foundations makes the distinction explicit.

2. Mathematical formulation

2.1 Objects, resources, and a reference measure

Let k be a positive size coordinate on a finite domain $D = [a, b]$, with $0 < a < b < \infty$. Choose a reference measure μ, with $0 < \mu(D) < \infty$, for the classes being compared. Let $n_\mu(k) = dN/d\mu$ denote object abundance and $\bar{q}(k) = E[q \mid k] > 0$ the conditional arithmetic mean of an additive resource per object. Then

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k) n_\mu(k). \tag{1}$$

For an observed class, the corresponding identity is simply $\sum_i q_i = N_{\text{class}} \bar{q}_{\text{class}}$. Resources may be a standing stock, such as biomass, or an accumulated event quantity over a declared observation period. The latter is not a stock resident in a size class, and neither is a flux through size classes. Additivity is needed for (1); a dynamical conservation law is not.

Define the **neutral allocation form of orthopolity relative to** μ, denoted O_μ, by

$$\mathcal{O}_\mu(k) = C > 0 \quad \text{for } \mu\text{-almost every } k \in D. \tag{2}$$

This is an assumption about the neutral allocation, rather than a claim that every constrained system has this profile. It gives $n_\mu = C/\bar{q}$. Equivalently, if an object is sampled uniformly from the population, its size density with respect to μ is

$$p_N(k) = \frac{1}{Z\bar{q}(k)}, \qquad Z = \int_D \frac{d\mu(k)}{\bar{q}(k)} < \infty. \tag{3}$$

Sampling instead in proportion to the resource held gives $p_R(k) = \bar{q}(k) p_N(k)/E_N[q] = 1/\mu(D)$. Thus uniform resource-weighted size and inverse-cost object abundance are equivalent statements, under the same reference measure. This is an accounting equivalence, not a derivation of the allocation assumption. The count, resource total, and constants also satisfy $N = CZ$ and $R = C\mu(D)$. For a finite observed population, the identity is exact for class totals; a smooth density is a population model or approximation.

Finite capacity and complete hierarchies. The box intuition has an exact formulation that precedes the neutral-allocation postulate. Let ν be a nonnegative additive resource measure on a system Ω, with $B = \nu(\Omega) \in (0, \infty)$. This physical resource measure differs from the class comparison measure μ. For disjoint units $A_i \subseteq \Omega$ with requirements $q_i = \nu(A_i)$, finite additivity gives the capacity bound

$$N_{\geq t} = \#\{i : q_i \geq t\} \leq B/t, \qquad t > 0.$$

Indeed, $tN_{\geq t} \leq \sum_{q_i \geq t} q_i \leq B$. Equal units of resource q therefore obey $Nq \leq B$, with equality when they exhaust the budget. In a cube of side L, cubes of side $\ell = L/m$, for integer $m \geq 1$,

tile it with $N = m^3$ and $N\ell^3 = L^3$. For other side lengths an aligned grid accommodates $\lfloor L/\ell \rfloor^3$ cubes; the volume bound remains $N\ell^3 \le L^3$. The grid construction is not a claim about optimal packing under arbitrary shape or orientation constraints.

Hierarchy theorem. Suppose each level j consists of a nonempty finite collection of positive-resource sets A_{ji}, and define the multiplicity $m_j(x) = \sum_i \mathbf{1}_{A_{ji}}(x)$. If N_j is their number and $\bar{q}_j = N_j^{-1} \sum_i \nu(A_{ji})$, then

$$N_j \bar{q}_j = \int_\Omega m_j(x)\, d\nu(x) = B\gamma_j, \qquad \gamma_j = B^{-1} \int_\Omega m_j \, d\nu.$$

This follows by integrating the finite sum of indicator functions. In particular, if every level completely partitions the same resource, $m_j = 1$ almost everywhere and

$$N_j \bar{q}_j = B, \qquad \frac{N_j}{N_h} = \frac{\bar{q}_h}{\bar{q}_j}.$$

No equal-size assumption is needed for the mean-resource relation. Nesting between levels is allowed: a unit cube, its eight half-side subcubes, and its sixty-four quarter-side subcubes each account for one unit of volume at their respective levels. Within-level disjointness permits the same matter to be represented at several resolutions. Summing J complete levels counts that underlying resource J times; it does not create J independent physical budgets. Hollow containers likewise require a choice between enclosed volume and wall material.

For incomplete disjoint levels, γ_j is the occupied resource fraction; for overlapping descriptions it is the mean multiplicity and may exceed one. The exact ratio becomes $N_j/N_h = (\gamma_j/\gamma_h)(\bar{q}_h/\bar{q}_j)$. Thus independently measured coverage supplies a quantitative correction to the inverse relation. A bound $m_j \le M$ also gives $N_{j,\ge t} \le MB/t$. Arbitrary overlap with no multiplicity bound has no finite capacity bound based on B alone. Complete partitions fix the comparison structurally: one full accounting per level. Equal steps in logarithmic size arise when each refinement uses the same scale factor.

This theorem establishes an exact inverse count–mean-resource relation across complete descriptions. A mixed population's size classes usually account for different parts of one budget. Writing their fractions as f_j gives $N_j \bar{q}_j = Bf_j$, with $\sum_j f_j \le 1$ for disjoint classes. Equal allocation across equally weighted classes then means equal f_j; finite capacity alone leaves those fractions undetermined. Sections 2.5 and 2.8 supply additional symmetry and dynamical conditions for that allocation statement. Additivity and multiplicity accounting are established mathematics (Tao, 2011); inverse-scale covering counts also have a developed geometric theory (Bishop and Peres, 2017). The capacity and hierarchy derivation gives proofs, examples, and the implications for composite resources.

2.2 Effective resources in systems with coupled requirements

An effective resource can combine several constituent requirements. For an object i, let $q_{i\ell}$ be its requirement or assigned amount of constituent ℓ, and let $s_\ell > 0$ be a fixed reference quantity in the same units. A general representation is

$$Q_i = Q_0 F\left(\frac{q_{i1}}{s_1}, \ldots, \frac{q_{im}}{s_m}; z_i, \theta\right) > 0, \qquad \bar{q}_{\text{eff}}(k) = E[Q_i \mid k].$$

Here Q_0 supplies effective-resource units, z_i records relevant environmental or historical conditions, and θ specifies the combination rule. The reference scales are declared independently of the observed abundance. Nonlinearity can represent complementarity, substitution, saturation, or a bottleneck. The arithmetic class mean is taken after combining the constituent resources.

In general, $E[F(X,Y) \mid k] \neq F(E[X \mid k], E[Y \mid k])$; for $F(X,Y) = XY$, the difference is the conditional covariance. Marginal resource means alone may therefore miss a dependence that changes the effective resource.

The quantity $R_{\text{eff}} = \sum_i Q_i$ is additive over the declared object partition, so equation (1) still holds with $\bar{q} = \bar{q}_{\text{eff}}$. Nonlinearity within an object does not prevent this additivity across objects. It does, however, make the physical partition important: $\sum_i F(\mathbf{q}_i)$ need not equal $F(\sum_i \mathbf{q}_i)$. Splitting or merging objects can change the effective total. Nor is R_{eff} automatically conserved merely because some constituent resources are conserved. The proposed combination must be interpreted as a physical stock, an accumulated requirement, an opportunity cost, or another explicitly defined quantity, with its appropriate balance law. Collective interaction terms must be included or assigned to objects through a physical accounting convention without double counting.

The hierarchy theorem yields a further restriction when an effective resource is required to describe the same underlying constituents at different resolutions. Let F be homogeneous of degree h, and split a parent constituent vector into r identical proportional parts. Their effective total is

$$\sum_{i=1}^{r} Q_0 F(\mathbf{x}/r) = r^{1-h} Q_0 F(\mathbf{x}).$$

Exact level invariance in this proportional subdivision therefore requires $h = 1$ for a positive parent and $r > 1$. For product resources this becomes $\sum_\ell \theta_\ell = 1$. Arbitrary heterogeneous subdivision is more restrictive: invariance requires $F(\mathbf{x} + \mathbf{y}) = F(\mathbf{x}) + F(\mathbf{y})$. If all nonnegative constituent vectors are admissible, F is finite and nonnegative, and $F(0) = 0$, additivity implies $F(\mathbf{x}) = \sum_\ell c_\ell x_\ell$ with $c_\ell \geq 0$. Thus nonlinear composition remains admissible on a physically specified object partition, or on a restricted family of constituent proportions, while invariance under every regrouping imposes a linear form. The supporting derivation proves these statements. They give a consistency test for applying a composite at several resolutions, without imposing homogeneity on resources defined only at one level.

For algae, nutrient stocks per cell and available nutrient budgets differ; light can be an absorbed photon flux or an exposure integrated over a specified interval. Space and water can be capacities or environmental conditions. Time can enter through residence time, growth history, or an additive occupancy such as cell-hours; elapsed time is not automatically a divisible stock. For cities, land, buildings and people are stocks, while food, water, energy and money can involve stocks, flows and exchanges over the chosen observation horizon. The combination rule must respect those distinctions and avoid counting a resource twice along a supply chain.

Multiple physical budgets remain explicit. For fixed class requirements, feasibility can require $\sum_j N_j q_{\ell j} \leq B_\ell$ for each constituent ℓ. A scalar Q does not generally replace all these inequalities. A specified optimization model can produce a local effective cost $Q_i = \sum_\ell \lambda_\ell q_{i\ell}$, with dimensionally compatible scarcity prices λ_ℓ, but those prices and their range of validity belong to that model (Boyd and Vandenberghe, 2004). A bottleneck gives another illustration: for a separately supported population of identical objects with fixed requirements, $N_{\max} = \min_\ell B_\ell / q_\ell$ depends nonlinearly on the budgets. It does not prescribe the allocation among competing types. These are ways to derive candidate effective quantities; neither scalarization nor feasibility by itself proves neutrality.

A simple nonlinear candidate is a product of positive normalized inputs, $Q = Q_0 \prod_\ell (q_\ell / s_\ell)^{\theta_\ell}$. If constituent requirements are deterministic functions $q_\ell(k) \propto k^{d_\ell}$, then $Q(k) \propto k^{d_{\text{eff}}}$ with $d_{\text{eff}} = \sum_\ell \theta_\ell d_\ell$. Under logarithmic neutrality the abundance density exponent is $1 + d_{\text{eff}}$. For differentiable F, the more general local elasticity is

$$\frac{d\ln Q}{d\ln k} = \sum_{\ell} \frac{\partial \ln F}{\partial \ln(q_\ell/s_\ell)} \frac{d\ln q_\ell}{d\ln k},$$

when the environment and parameters are fixed and F has no further explicit size dependence. Varying constituent elasticities or interactions can generate curvature. This chain rule for specified functions is distinct from a claim that independent random resource distributions have additive tail exponents. With heterogeneous objects, the class mean $E[Q \mid k]$ remains the relevant quantity. The effective-resource derivation develops the finite-bin and heterogeneity implications.

2.3 Inverse identification and Zipf examples

The law can be used in two directions. Given a resource model and a specified constraint, it predicts abundance. Given an abundance profile, it supplies information about the effective resource required by that allocation. Write a positive constrained profile as

$$\bar{q}_{\text{eff}}(k) n_\mu(k) = C a(k), \qquad a(k) > 0,$$

where neutrality has $a = 1$. The class-exchange example below derives $a = e^{-V}$ from prescribed kinetics. In other systems a needs its own mechanism or conditional assumption; an arbitrary fitted factor is not an explanation. On a domain with positive population density, inversion gives

$$\frac{\bar{q}_{\text{eff}}(k)}{a(k)} = \frac{C}{n_\mu(k)}, \qquad \frac{\bar{q}_{\text{eff}}(k)/a(k)}{\bar{q}_{\text{eff}}(k_0)/a(k_0)} = \frac{n_\mu(k_0)}{n_\mu(k)}.$$

Thus a distribution identifies the shape of effective resource divided by its constraint factor, conditional on the comparison measure. With the constraint profile and its normalization specified, an independently measured effective-resource total fixes $C = R_{\text{eff}}/\int_D a\, d\mu$. Without further information, the constituents and their combination are not unique; multiplying both $\bar{q}_{\text{eff}}$ and a by the same positive size-dependent function leaves the prediction unchanged. This is a precise statement of the inverse information available, not a reason to discard inverse inference. Measured constituents, restricted combination families, and responses in different environments can resolve parts of that ambiguity. Finite observed counts require a sampling model and bin integration; a zero-count bin does not establish infinite physical cost.

A constructive inverse example. Prespecify $Q = Q_0 X^{\theta_1} Y^{\theta_2}$ for positive normalized resources. In environment s, suppose independently measured class requirements scale as $X_s \propto k^{d_{1s}}$ and $Y_s \propto k^{d_{2s}}$, with no within-class variation, and a known constraint has $V_s = v_s \ln(k/k_0)$. A Pareto count density $dN/dk \propto k^{-(1+\alpha_s)}$ then implies

$$\alpha_s - v_s = \theta_1 d_{1s} + \theta_2 d_{2s}.$$

Two environments with scaling vectors $(1,1)$ and $(1,2)$ and corrected exponents $\alpha_s - v_s$ equal to 1.5 and 2 identify $\theta_1 = 1$ and $\theta_2 = 1/2$. The inferred nonlinear resource is $Q = Q_0 X\sqrt{Y}$. A third environment with scaling vector $(2,1)$ predicts $\alpha_3 = 2.5 + v_3$ without refitting. The two independent scaling directions identify the composition within this family; the third supplies a test. The remaining scale Q_0 requires an independent anchor. These are algebraic illustrative values, not measurements. They show how allocation distributions can reveal a composite resource and generate further predictions.

Identification and transfer. More generally, assume a common product composition $Q = Q_0 \prod_{\ell=1}^{m} X_\ell^{\theta_\ell}$, a fixed logarithmic comparison measure, independently characterized constituent exponents $D_{s\ell}$, and known constraint slopes v_s. Assume deterministic class requirements or write $X_{s\ell} = A_{s\ell}(k/k_0)^{D_{s\ell}} U_{s\ell}$ with a finite positive residual moment $E[\prod_\ell U_{s\ell}^{\theta_\ell} \mid k, s]$ independent of size. Then the corrected abundance slopes satisfy

$$b = D\theta, \qquad b_s = \alpha_s - v_s.$$

For a compatible b and unrestricted real exponents, θ is unique precisely when D has full column rank. With deficient rank, the compatible parameters are $D^{+}b + \ker D$, provided b lies in the column space. A new environment with constituent row d_* still has an identified prediction if and only if d_* lies in the row space of D, for unrestricted real exponents. Restrictions such as nonnegative exponents can reduce this ambiguity further; they must be included in the identification problem.

Every vector ℓ satisfying $\ell^{\top} D = 0$ supplies a parameter-free cross-environment consequence,

$$\ell^{\top} b = 0.$$

For the three environments above, this gives $b_3 = 3b_1 - b_2$. The third slope is predicted before its abundance data are used. Thus additional environments can test a shared effective resource even when an individual profile supplies only a conditional reconstruction.

Proof and precision. Two parameter vectors give the same calibration slopes exactly when their difference lies in $\ker D$. A target row gives the same prediction for all such vectors exactly when it annihilates that kernel. Multiplication by a left-null vector gives the stated contrast. For full column rank and fixed D, least-squares recovery obeys

$$\|\widehat{\theta} - \theta\|_2 \le \frac{\|\delta b\|_2}{\sigma_{\min}(D)}.$$

Near-proportional constituent scalings can therefore make a formally unique composition practically unstable. A contrast with slope-error covariance Σ_b has variance $\ell^{\top}\Sigma_b \ell$, including correlations between environments. Uncertain constituent scalings, constraint corrections, and size-dependent heterogeneity require their own error model; this fixed-design bound does not cover them. These statements apply standard linear inverse-problem results to the allocation model. The supporting identification analysis supplies generalized least squares, partial identification, and reproducible synthetic recovery and transfer checks.

City sizes. Upper city-size tails often admit Pareto descriptions, with Zipf scaling as a prominent special case. The result depends on how cities and the fitted size range are defined. Eeckhout (2004) found a lognormal description across the full US Census Places distribution; Ioannides and Skouras (2013) found a lognormal body and Pareto tail, with an estimated tail exponent of 1.25 for their preferred Census Places model. These findings concern different parts and definitions of an urban system, not a requirement that every city-size observation follow exact Zipf scaling.

Let S denote population per city and $n(S)$ the positive number density of cities per unit population interval. For a Pareto survival exponent α, $N(\ge S) \propto S^{-\alpha}$ and $n(S) \propto S^{-(1+\alpha)}$; ranked size scales as $S(r) \propto r^{-1/\alpha}$. Finite upper cutoffs modify the survival and rank relations near the largest city. Under logarithmic allocation,

$$\frac{dN}{d\ln S} = Sn(S), \qquad \frac{\bar{q}_{\mathrm{eff}}(S)}{a(S)} \propto S^{\alpha}.$$

Exact Zipf scaling has $\alpha = 1$. If $a = 1$ and population is the resource, $q(S) = S$, the relation expresses equal population in equal logarithmic city-size intervals. If the operative resource is a nonlinear combination of urban requirements, the same observation instead constrains that combination to have approximately linear net scaling, after accounting for a. This is the proposed orthopolity interpretation and an inverse target for measurements of land, infrastructure, energy, water and other inputs. The exponent alone does not identify the combination or assign a probability to its causal explanation. Existing proportional-growth mechanisms, including

Gabaix's (1999) model with lower-size regularization, provide comparisons that the resource account should explain, connect to, or distinguish. No new city dataset is analyzed here.

Language. A word type is a distinct lexical form and a token is one occurrence. For type w, let f_w be its token frequency and L_w its letter count. The corpus expenditure of letters on that type is $f_w L_w$. Equality on a counting measure over word types would therefore imply $f_w \propto L_w^{-1}$. Zipf's rank-frequency relation $f(r) \propto r^{-\zeta}$ concerns a different coordinate. In a smooth rank approximation, letter expenditure per logarithmic rank is $rf(r)L(r) \propto r^{1-\zeta}L(r)$. For $\zeta = 1$, tokens per log rank are constant, while letters per log rank vary with word length. Finite rank groups require the actual sums, including tied frequencies. Length classes introduce another distinction: if M_L vocabulary types have length L, equal letter expenditure per type gives total token count at that length proportional to M_L/L. The number of available types is part of the comparison measure; a peaked word-length distribution alone does not negate this conditional allocation.

The abbreviation tendency, in which frequent words are shorter, motivates resource candidates without fixing either exponent or comparison measure. Contextual information is also relevant to word length (Piantadosi, Tily and Gibson, 2011). A declared candidate is $Q_w = Q_0 L_w^{\theta_L} P_w^{\theta_P}$, where P_w is a dictionary phoneme count. Under neutral type allocation it predicts $p_\theta(w) \propto L_w^{-\theta_L} P_w^{-\theta_P}$ on a fixed vocabulary. Section 5 applies this inverse model across text genres. Letter and phoneme counts are symbolic resource candidates; sonority, duration and acoustic energy need separate measurements and constitutive definitions.

2.4 The measure changes the exponent

Write $u = \ln(k/k_0)$ for a fixed reference size k_0. For $\bar{q}(k) = q_0(k/k_0)^d$, with $d > 0$, two different hypotheses give:

Equal resource with respect to	**Resource spectrum**	**Count density with respect to** dk
Linear size, $d\mu = dk$	$\bar{q}(k)dN/dk = C$	$dN/dk \propto k^{-d}$
Logarithmic size, $d\mu = du = dk/k$	$\bar{q}(k)dN/du = C$	$dN/dk \propto k^{-(d+1)}$

The second row follows because $dN/du = k\,dN/dk$. For biomass with $q = k = m$, the predictions are consequently density exponents 1 and 2, respectively. The 2017 manuscript's explicit uniform density over a linear wealth interval is not interchangeable with the logarithmic convention used in the empirical analyses below. Adopting logarithmic classes is a substantive specification, motivated by size-spectrum practice.

Logarithmic allocation is unchanged by changing size units or logarithm base, apart from a constant density multiplier. It also survives $z = (k/k_0)^c$ for constant $c > 0$, since $d\ln z = c\,d\ln k$. It is not invariant under arbitrary nonlinear coordinates. In general, $dR/dv = (dR/du)|du/dv|$. Choosing a transformation after observing R can therefore manufacture flatness; the coordinate and measure must be fixed independently. More precisely, changing coordinates while carrying the same measure along preserves the allocation statement. Replacing that measure by a uniform measure in the new coordinate changes the hypothesis. The complete hierarchy in Section 2.1 chooses its levels through the construction of the system; that structural choice is separate from relabeling those levels.

Under $O_{\log}$ and the power-cost assumption, the slope of abundance per log interval is $-d$, the ordinary density exponent is $d + 1$, and the survival function on $[a, b]$ is

$$P(K \geq k) = \frac{k^{-d} - b^{-d}}{a^{-d} - b^{-d}}. \tag{4}$$

The familiar survival exponent d and rank-size exponent $1/d$ are approximations away from the upper boundary, not exact finite-domain identities. For an unbounded domain with $d > 0$, count can be finite while logarithmically uniform resource has infinite total. Finite boundaries are essential to a finite-budget interpretation.

2.5 Allocation postulates and limits of the accounting

Resource symmetry as a candidate law. The proposed rationale is that, absent restrictions favouring particular resource costs, allocation should not distinguish those costs. A natural law can adopt this as a physical postulate without a deeper mechanistic derivation. One precise sufficient formulation uses K declared classes with fixed positive per-object costs q_i, allocations $R_i = q_i N_i$, and fixed total $\sum_i R_i = R > 0$. If the probability law of $(R_1, \ldots, R_K)$ is invariant under exchanging allocations across the fixed classes, then symmetry and the sum give

$$E[R_i] = R/K, \qquad E[N_i] = R/(Kq_i).$$

This postulate selects resource allocations as the symmetric quantities; symmetry of counts would give a different prediction. It is stronger than conservation and is not implied by an unspecified absence of constraints. Indivisible objects may require an approximation where exact exchangeability is infeasible. The class measure also matters: with cost itself as coordinate, equal resource per dq predicts $dE[N]/dq \propto q^{-1}$, whereas equal resource per $d\ln q$ predicts $dE[N]/dq \propto q^{-2}$. Transformation-based treatments of indifference likewise require a specified symmetry (Jaynes, 1968); an inferential rule does not itself establish a physical law.

Equal expectation does not establish equal snapshots or relaxation toward equality. In a divisible-resource idealization, assigning all resource to one uniformly chosen class has the required symmetry and unequal allocation in every realization. Time-average equality needs an additional condition connecting time averages to expectations or its own empirical postulate. A test must therefore specify what "tends to allocate" means and identify eligible systems through conditions independent of their observed allocation. A deeper mechanism is optional for an empirical law; the scope and predictive evidence are not.

Conservation does not imply equal allocation. On any bounded logarithmic domain, $\mathcal{O}(u) = A\exp(\beta u)$ can be normalized to the same resource total for every β. Conservation fixes the integral and leaves its shape undetermined. Likewise, maximum entropy over finitely many equally weighted object classes, constrained by a mean cost, yields $p_j \propto \exp(-\lambda q_j)$. Different reference measures or constraints give different distributions (Jaynes, 1957; Visser, 2013).

Equipartition does not imply a power law without a cost law. If $\bar{q}(k) = q_0 \exp(k/k_0)$, then $O_{\log}$ gives $dN/du \propto \exp(-k/k_0)$ and $dN/dk \propto \exp(-k/k_0)/k$. Neither is a power law. Geometric dimension supplies d only in a specified geometric model such as similar objects with volume proportional to length cubed. It does not establish the number of independent resource inputs in an arbitrary system.

Independent inputs do not generally add tail exponents. For example, two independent unit-scale Pareto variables with densities hx^{-h-1}, $x \geq 1$, $h > 0$, have product density $h^2 z^{-h-1} \ln z$, $z \geq 1$. Integration over the possible factorizations introduces a logarithmic factor. This limits the independent-input argument in the 2017 essay.

Forward tests and inverse inference have different roles. For any positive abundance, defining $\bar{q} = C/n_\mu$ makes (2) hold identically. In Section 2.3 this is the starting inverse relation: it identifies a conditional effective-resource profile and motivates resource hypotheses.

Agreement on the same fitted profile is not independent evidence for the law or for the inferred physical composition. A forward test fixes the resource model through definitions, independent measurements, or a separate calibration, then checks a prediction on additional observations. A constraint inferred from the original profile likewise needs additional information or a transferred prediction to establish its explanatory value.

2.6 What an ensemble statement would mean

Suppose a system has a power-shaped resource spectrum $\mathcal{O}_i(u) = A_i \exp(\beta_i u)$. A proposed ensemble condition $E[\beta_i] = 0$ concerns mean slopes. It does not imply $E[\mathcal{O}_i(u)]$ is constant. With equal $A_i = A$ and $\beta_i \sim N(0, \tau^2)$,

$$E[\mathcal{O}_i(u)] = A \exp(\tau^2 u^2/2), \tag{5}$$

which is curved whenever $\tau > 0$. Equal-total normalization changes this formula but does not make mean-zero slopes sufficient: averaging normalized spectra proportional to $\exp(cu)$ and $\exp(-cu)$ on a symmetric domain produces a nonconstant hyperbolic cosine. A median slope is a different estimand again. A claim about average resource occupancy requires spectra, a common domain, and an explicit weighting of systems; slopes alone do not identify it.

More generally, let $H(u) = \sum_{i=1}^{n} w_i A_i e^{\beta_i u}$ on a common log-domain, with fixed positive weights and amplitudes. Then

$$H''(u) = \sum_i w_i A_i \beta_i^2 e^{\beta_i u} \geq 0.$$

If any slope is nonzero, the derivative is strictly positive, so the arithmetic mixture cannot be constant on an open interval. Exact flatness requires every contributing slope to be zero. With $p_i(u) = w_i A_i e^{\beta_i u}/H(u)$, its log-curvature is $d^2 \ln H/du^2 = \mathrm{Var}_{p(u)}(\beta)$. Per-system total normalization preserves these conclusions. They are standard convexity consequences for exact power spectra; arbitrary curved profiles can cancel to a flat mean. They therefore neither exclude a general resource-symmetry law nor convert noisy fitted slopes into a profile test. The scientific assessment gives a quantitative bound for approximate flatness and its weighting restrictions.

Mean and variance also do not require a Gaussian distribution. Gaussianity is an additional maximum-entropy modelling choice, and selecting it by AIC is not a goodness-of-fit test. If $\beta_i \sim N(\mu_\beta, \tau^2)$ and reported slope error is independent $N(0, \sigma_i^2)$, the predicted probability that the *reported slope* lies in $[-c_i, c_i]$ is

$$\Phi_{\mathrm{N}}\left(\frac{c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right) - \Phi_{\mathrm{N}}\left(\frac{-c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right). \tag{6}$$

The expression $2\Phi_{\mathrm{N}}(c_i/\tau) - 1$ applies to zero-centred *latent* slopes. Comparing it to observed estimates without their errors is a different calculation. Fitting τ on the same slopes makes either comparison an internal diagnostic, not independent validation.

There is a valid simultaneous-resource restriction for an individual system: $\mathcal{O}_1/\mathcal{O}_2 = \bar{q}_1/\bar{q}_2$. Both spectra can be exactly flat only if their mean resource ratio is size-independent. Across systems, however, $\beta_{1i} - \beta_{2i} = d_{1i} - d_{2i}$ for power cost laws. Equal variances and correlation one follow only if that difference is constant and slope variance is nonzero, whether or not equipartition holds. They are not a general test of ensemble equipartition.

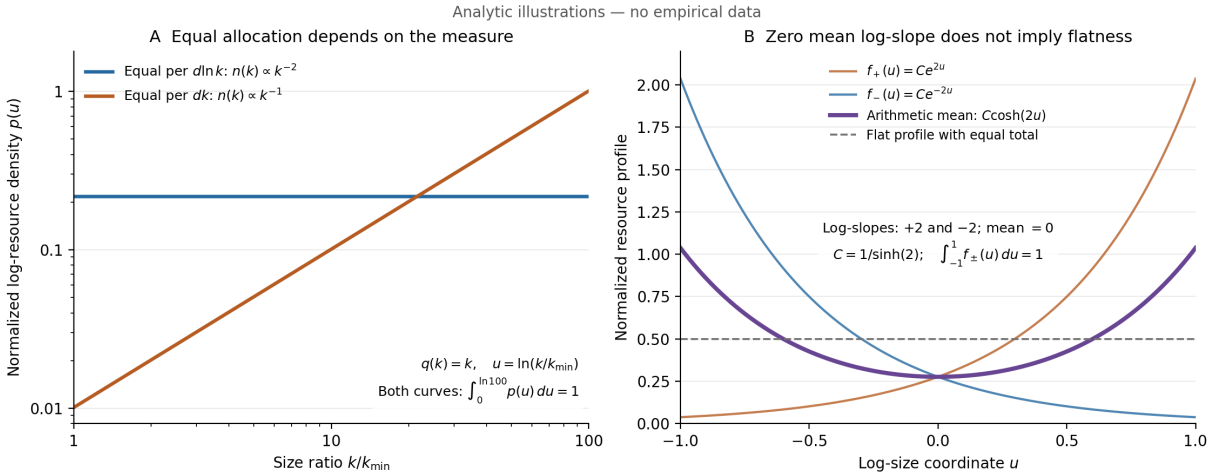


Figure 1. Analytic illustrations, not empirical data. (A) With resource equal to size, linear and logarithmic resource equipartition produce different count exponents. Both curves show unit-total resource density in the logarithmic coordinate. (B) Two unit-total resource profiles on a common domain have opposite log-slopes and therefore zero mean log-slope, but their arithmetic mean is not flat. Generated by run_theory.py; a vector version is available.

2.7 System constraints and observable distributions

Three levels of description must be kept distinct: an underlying resource-equalizing tendency, the neutral signature (2), and the outcome of the complete system. The latter depends on the proposed tendency together with resource costs, other processes, forcing, and initial and boundary conditions. A model containing an equalizing contribution can therefore predict a non-neutral stationary state or a transient with no clear neutral signature. Establishing such a contribution requires an explicit model or discriminating observations; equation (1) alone does not identify it.

Age and height distributions are motivating examples of constrained outcomes. A worked age-distribution account must specify birth history, survival, and migration; a height account must specify developmental processes, biological variation, and environmental conditions, as well as the population sampled. An orthopolity application must additionally identify the resource, its concentrations, and the comparison measure. No completed orthopolity model of either example is supplied here. The relevant task is to derive the combined outcome and explain how its form depends on the constraints.

A failure of the neutral prediction is a finding about that prediction in the stated setting. It does not automatically establish the absence of the broader tendency, and counting distributions without its visible signature does not establish a general absence of evidence in Nature. A constrained model has its own predictions, which can succeed or fail. Specify the constraint through physical reasoning or independent measurements, derive its effect on the profile, and test the predicted response when it changes. Where recovery of neutrality is predicted, specify its conditions and timescale. This permits both neutral and non-neutral outcomes to inform the theory, while preserving the distinction between an observed departure, a proposed cause, and a tested explanation.

2.8 A mechanism on resource-concentration classes

Consider K fixed size classes, with positive reference widths w_i, fixed per-object resource costs q_i, nonnegative class stocks R_i, and conserved total $B = \sum_i R_i$. The classes represent populations of resource-bearing objects; the state is their aggregate resource. Equivalent counts $N_i = R_i/q_i$

may be continuous or expected counts, and their total need not be conserved. This effective model does not specify individual-object assembly or demographic kinetics. The full derivation states its interpretation and assumptions.

Let $g_{ij} = g_{ji} \geq 0$ be conductances on a connected undirected graph. Neutral exchange has current $J^0_{ij} = g_{ij}(R_i/w_i - R_j/w_j)$ and dynamics $\dot{R}_i = -\sum_j J^0_{ij}$. Positivity follows from nonnegative jump rates g_{ij}/w_i; antisymmetry of currents conserves total resource. Connectedness makes the stationary resource density constant, giving

$$R^0_i = \frac{Bw_i}{\sum_j w_j}, \qquad N^0_i = \frac{Bw_i}{q_i \sum_j w_j}.$$

This is a mechanism selecting equal resource per declared class measure. The local exchange symmetry and the measure are assumptions of the model; they are not supplied by conservation alone. To model a constraint, prescribe dimensionless class preferences V_i and add downhill transitions:

$$a_{ij} = \frac{g_{ij}}{w_i} \exp(\max\{V_i - V_j, 0\}) .$$

The rate a_{ij} is from i to j. Its neutral part g_{ij}/w_i remains, and the additional part is nonnegative. With column generator $A_{ji} = a_{ij}$ for $j \neq i$, $A_{ii} = -\sum_{j \neq i} a_{ij}$, define

$$\pi_i = \frac{w_i e^{-V_i}}{Z}, \qquad Z = \sum_j w_j e^{-V_j}.$$

Then $\pi_i a_{ij} = \pi_j a_{ji}$ on every edge. Detailed balance selects the unique stationary resource $R^V_i = B\pi_i$. This uses standard reversible-chain mathematics (Levin and Peres, 2017), specialized here to fixed concentration classes and an explicit additive constraint mechanism. At that stationary state the neutral edge current is

$$J^0_{ij} = \frac{Bg_{ij}}{Z}(e^{-V_i} - e^{-V_j}), \qquad J^c_{ij} = -J^0_{ij}.$$

Thus a nonuniform stationary distribution can have an active equalizing contribution balanced by the constraint. Setting V to a constant removes the added rates and lets the actual constrained allocation relax toward neutrality.

For $p = R/B$, set $E^2 = \sum_i (p_i - \pi_i)^2/\pi_i$. Writing $z_i = p_i/\pi_i$ gives

$$\frac{dE^2}{dt} = -2\sum_{i<j} \pi_i a_{ij}(z_i - z_j)^2 \leq -2\lambda E^2,$$

where $\lambda > 0$ is the smallest nonzero eigenvalue of the symmetric matrix $-D_\pi^{-1/2} A D_\pi^{1/2}$. Consequently $E(t) \leq E(0)e^{-\lambda t}$ and $\mathrm{TV}(p, \pi) \leq E/2$. The same bound applies after release, using the neutral generator and the state actually present at the switch. If connections are removed, each component retains its own resource total and only component-wise equalization follows.

The simulation fixes 24 equal-log size classes on $[1, 64]$, $q_i = k_i^{1.5}$, and adjacent conductances $1/\Delta \ln k$. A prescribed quadratic preference in $\ln k$, centered at $k = 8$ with width 0.75, produces a curved resource profile. The corresponding continuous count form is $dN/dk \propto k^{-2.5} e^{-V(\ln k)}$, a truncated lognormal shape for this quadratic choice. Each phase runs for eight relaxation times. The neutral and constrained spectral gaps are 0.569805 and 1.991712 per model time unit. The final relative errors in the neutral, constrained, and released phases are 4.73×10^{-4}, 2.41×10^{-7}, and 5.51×10^{-11}, each below its predicted bound. The opposing stationary drift vectors cancel to Euclidean residual 3.22×10^{-15}.

There are 4,096 equal-resource packets in each of 128 realizations, initially all in the first class. Exact matrix-exponential checkpoint transitions preserve the state at both switches. Packets remain independent, giving class-count law $X(t) \sim \text{Multinomial}(4096, p(t))$ and share covariance $(\text{diag}(p) - pp^T)/4096$. Uneven finite snapshots are therefore predicted even at neutral mean allocation. Rates, costs, preferences, and predictions were fixed before drawing the stochastic trajectories. The computational results and checks are retained with the model report; the simulation supplies a constructive example, not new natural-system observations.

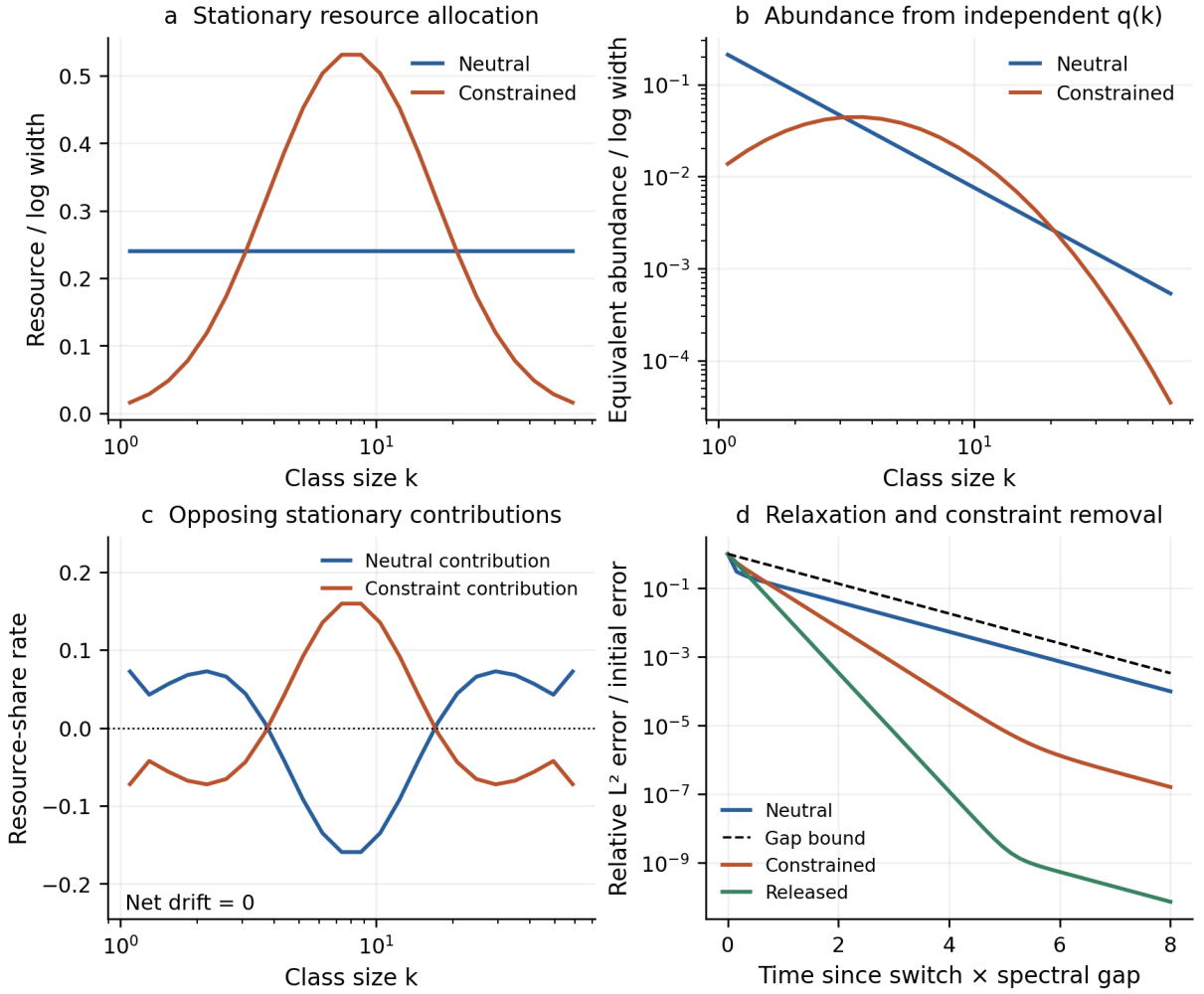


Figure 2. Fixed class-exchange model. Equalizing kinetics select neutral resource shares; an added class preference selects a curved profile and changes equivalent abundance. At constrained equilibrium the neutral and added contributions cancel. Removing the preference preserves the current state and predicts recovery under the neutral dynamics. The displayed convergence curves are deterministic expectations; stochastic packet realizations and their uncertainty are retained in the linked report.

2.9 A physical realization from electromagnetic transport

For electromagnetic energy density u and Poynting vector $\mathbf{S}$, Maxwell's equations give $\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$ (Poynting, 1884; Haus and Melcher, 1989). Define $P(r) = \int_{S_r} \mathbf{S} \cdot \mathbf{n}\, dA$ outside a source enclosed by an inner sphere r_0 with steady power $P(r_0) = L$. Integrating the balance over a transparent, source-free shell with no net energy accumulation gives $P(r_2) = P(r_1) = L$. Isotropy is a separate condition: it makes the local radial flux density $F = L/(4\pi r^2)$; otherwise this is its sphere average.

Under outgoing radial transport at constant speed c, without scattering or secondary emission in the modeled component, a specified absorber with coefficient $\kappa(r)$ gives

$$\frac{dP}{dr} = -\kappa(r)P, \qquad P(r) = L\exp\left[-\int_{r_0}^{r}\kappa(s)\,ds\right].$$

This is the absorption law specialized to spherical transport (Condon and Ransom, 2016). An absorbing layer with optical depth $\ln 2$ halves downstream power. The missing radiative power is deposited in matter, so the modeled radiation component has an open budget. Its departure follows from the prescribed medium and the same energy balance.

These nested surfaces receive the passage of the same energy; their powers cannot be summed as disjoint resource shares. Disjoint shell stocks do give an additive comparison: in transparent stationary radial transport, $dU = (L/c)dr$. Thus equal radial thicknesses hold equal energy, whereas $dU/d\ln r = Lr/c$ increases with radius. The resource, units, and comparison measure determine which equality is physically meaningful.

The response to removing an absorber is also determined. For sphere-averaged u, with $e = 4\pi r^2 u = P/c$, the transport equation is $\partial_t e + c\partial_r e = -c\kappa(r,t)e$. Suppose the source remains constant, the initial field is stationary under $\kappa_{\text{old}}(r)$, and absorption is set to zero throughout the modeled domain at $t = 0$. For $t \geq 0$, characteristics give

$$P(r,t) = L\exp\left[-\int_{r_0}^{\max\{r_0,r-ct\}}\kappa_{\text{old}}(s)\,ds\right].$$

Every radius recovers by $(r - r_0)/c$. Downstream of a layer $[a, b]$, the response starts at $(r - b)/c$ and completes at $(r - a)/c$. This is an idealized change of the absorption coefficient, with no radiation injected by the removal process. The physical derivation states the boundary history and all transport assumptions. It supplies an established physical realization of a declared equality and its constrained departure; it does not equate radial shells with organism size classes or derive abundance laws from an inverse-square flux.

2.10 Classical equipartition and quantum radiation

Thermal electromagnetic radiation supplies a second physical realization with a different reference measure: counting field modes, including polarization. For a finite set of classical normal modes in canonical equilibrium at $T > 0$, each has two quadratic terms, $H_j = P_j^2/(2m_j) + m_j\omega_j^2 Q_j^2/2$. Gaussian integration over both coordinates gives $Z_j^{\text{cl}} \propto (\beta\omega_j)^{-1}$, with $\beta = (k_B T)^{-1}$, and therefore

$$\langle H_j\rangle = -\partial_\beta \ln Z_j^{\text{cl}} = k_B T.$$

A class of M_i modes consequently has mean energy $M_i k_B T$: the resource density is equal per mode, with independently defined weight $w_i = M_i$. This is equality of equilibrium means, not identical instantaneous energies. Equilibrium preparation or thermal coupling must be justified; uncoupled harmonic modes do not thermalize each other merely because their canonical distribution can be written down. This argument uses standard equipartition and supplies no relaxation time (Tong, n.d.).

Quantization changes the allowed states. Define the resource as thermal excitation energy above the ground state, excluding the temperature-independent zero-point term. For photon absorption and emission in thermal equilibrium, photon chemical potential is zero and a mode has excitation levels $\epsilon_n = nh\nu$. The geometric sum gives

$$Z_\nu = \frac{1}{1-e^{-x}}, \qquad \bar{\epsilon}_\nu = \frac{h\nu}{e^x - 1} = k_B T f(x), \qquad x = \frac{h\nu}{k_B T}, \quad f(x) = \frac{x}{e^x - 1}.$$

For $x > 0$, f decreases strictly from 1 toward 0. Its limits are $f(x) = 1 - x/2 + x^2/12 + O(x^4)$ near zero and $f(x) \sim xe^{-x}$ at large x. The quantum level spacing relative to thermal energy specifies the departure before any spectrum is observed. Raising temperature or lowering frequency recovers classical relative equality. This is a restriction of the available energy states; it does not establish the additive decomposition into opposed classical currents proved in Section 2.8.

In a large three-dimensional vacuum cavity, counting two polarizations gives the mode density $dM/(V d\nu) = 8\pi\nu^2/c^3$. Multiplication by $\bar{\epsilon}_\nu$ yields the Planck energy density and, for isotropic radiation, the spectral radiance:

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3(e^{h\nu/(k_B T)} - 1)}, \qquad B_\nu(T) = \frac{c}{4\pi} u_\nu(T).$$

Thus the classical equality per mode gives $u_\nu^{\mathrm{cl}} \propto \nu^2$ and $dU/(V d\ln\nu) \propto \nu^3$, neither a flat linear-frequency nor a flat log-frequency spectrum. If photons are the counted objects, $q(\nu) = h\nu$ and their mean number per mode approaches $k_B T/(h\nu)$ in the classical regime. Extending classical equipartition to infinitely many modes gives divergent total energy; quantization makes it finite, $U/V = 8\pi^5 k_B^4 T^4/(15h^3c^3)$. These are established results (Planck, 1901; Tong, n.d.; Condon and Ransom, 2016), here specifying the resource, comparison measure, and constraint in the allocation framework.

2.11 Connection to the measured thermal spectrum

FIRAS measured the cosmic microwave background against calibrated blackbody references and separated monopole, dipole, and Galactic components (Fixsen et al., 1996). NASA LAMBDA's distributed monopole table contains a 2.725 K blackbody plus the residuals from that published analysis, with marginal uncertainties and a Galactic model column. The residuals already follow temperature and foreground fitting. The table therefore supports a transparent re-expression of the published result; agreement with its embedded template cannot be counted as a new independent confirmation.

Our retrospective calculation retains all 43 rows and fixes $T_0 = 2.725$ K from the product definition. With wavenumber $\tilde{\nu}$ in inverse centimetres, $\nu = 100c\tilde{\nu}$. Radiance remains per hertz, with $1\,\mathrm{MJy/sr} = 10^{-20}\,\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$. Dividing the inferred energy density by the independent mode density gives

$$\frac{\widehat{\epsilon}_\nu}{k_B T_0} = \frac{c^2 I_\nu}{2\nu^2 k_B T_0} = \frac{I_\nu}{B_\nu^{\mathrm{RJ}}(T_0)}, \qquad B_\nu^{\mathrm{RJ}} = \frac{2\nu^2 k_B T_0}{c^2}.$$

The channels span 68.05–639.46 GHz and $x = 1.199$–11.262. Their transformed mode energies range from 0.518 to 1.32×10^{-4} of $k_B T_0$ (Figure 3). They illustrate the quantum departure, not the $x \ll 1$ classical plateau. The CMB can retain a Planck spectrum through collisionless cosmological redshifting; this example does not require current thermal contact with surrounding matter (Condon and Ransom, 2016, Section 2.6.2).

For residual diagnostics we use the source's original residual column, not the difference between rounded monopole intensities and a newly calculated curve. The source's approximate frequency covariance is $C_{ij} = \sigma_i \sigma_j Q_{|i-j|}$, using its 43 published lag coefficients (Fixsen et al., 1996, Section 3.3). The matrix is positive definite. Applying the linear radiance-to-mode conversion to both axes propagates this covariance to the mode-energy ratios. The residual quadratic form is $r^T C^{-1} r = 49.774$; it is a descriptive audit of the published approximation, with no new fit or goodness-of-fit probability. It need not reproduce the original full-analysis statistic, 46 for 40 degrees of freedom. Diagonal inverse-variance weighting gives residual RMS 18.90 kJy/sr, or

49.28 parts per million of the tabulated peak radiance. This defined amplitude summary does not assume that the channels are independent.

Using exact SI constants at the rounded source coordinates gives a discrepancy $I_{\mathrm{posted}} - [B_\nu(T_0) + r_{\mathrm{published}}]$ of -0.127 to $+7.598$ kJy/sr, with radiances in common units. This discrepancy is retained without assigning a cause or replacing the original residuals. Absolute calibration, foreground modeling, and the prior fits remain inherited from the published analysis. The thermal-radiation report retains the full derivation, source bytes, covariance transcription, and reproducible calculation. This example connects a physical equality and an independently derived quantum departure to established observational evidence under its documented conditions.

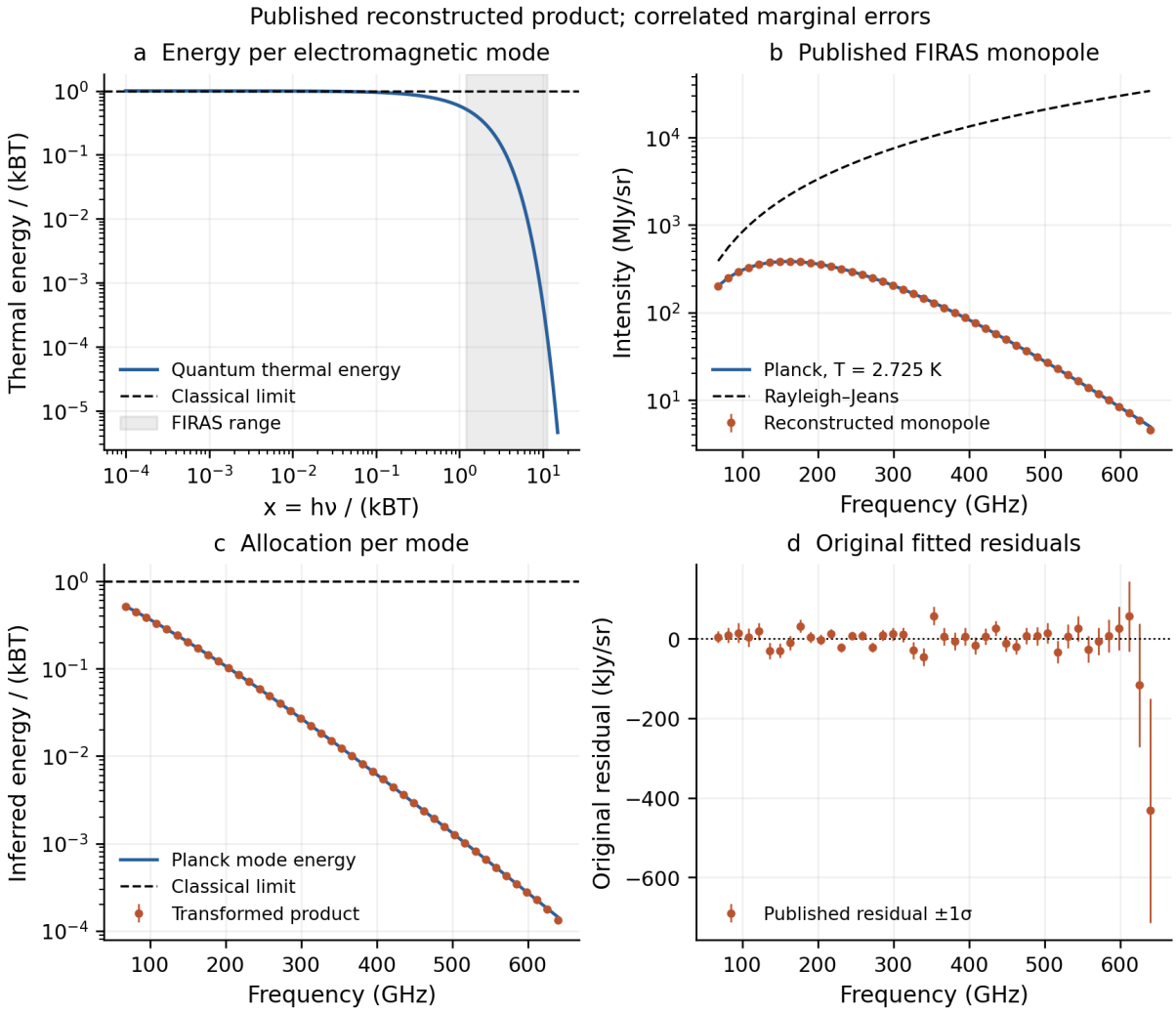


Figure 3. Thermal energy per electromagnetic mode. (A) The theoretical quantum factor $f(x)$ approaches classical equality as $x \to 0$. (B) The NASA FIRAS reconstructed monopole product and Planck and Rayleigh–Jeans radiances at the product's 2.725 K reference temperature. (C) The same product expressed as energy per mode relative to $k_B T_0$. (D) Original published residuals with marginal one-sigma uncertainties; channels are correlated. The Planck template is part of the distributed product, so panels B and C illustrate the published result rather than provide a new independent model test. All 43 channels are retained.

2.12 Growth, division, and the observation of resource allocation

A biological mechanism makes the comparison measure and a nonconstant constraint profile explicit. Let additive cell mass x grow as $\dot{x} = gx$, with common deterministic g, conservative

symmetric binary division, and no death. Assume a balanced asynchronous population with finite mean mass and a stationary chronological lineage following physiologically equivalent daughters. Established growth-fragmentation theory gives

$$p_L(x) = \frac{x p_P(x)}{E_P[x]}.$$

The mass-weighted population size distribution equals the chronological lineage size distribution (Genthon, 2022). This is a known sampling transformation, not a new cell-division law. For lineage birth and division sizes B, D, stationarity implies $D \overset{d}{=} 2B$. Monotone growth gives the fraction of cycles crossing x,

$$H(x) = P(B \le x < D) = F_B(x) - F_B(x/2).$$

Each crossing occupies time $dx/(gx)$; mean cycle duration is $\ln 2/g$. Therefore

$$p_L(x) = \frac{H(x)}{x \ln 2}, \qquad p_P(x) = \frac{2H(x)}{E[B^{-1}]x^2}, \qquad \frac{x^2 p_P(x)}{E_P[x]} = \frac{H(x)}{\ln 2}.$$

The last expression is the normalized mass share per logarithmic size. Independently measured birth/division events specify its envelope, instead of defining a constraint from the population residual. Every interval crossed by all cycles has $H = 1$ and equal mass per equal logarithmic width. Fixed birth mass b yields the familiar inverse-square density $2b/x^2$ on $[b, 2b)$; variable birth sizes generally remove an exact plateau. The framework thus predicts both the neutral form and an event-determined departure.

These conditions matter: growth noise changes the sampling transformation; asymmetric division requires mass-biased daughter tracking; old-pole retention need not sample equivalent physiology. Perfectly synchronized deterministic cycles need not converge to a balanced phase distribution. The derivation and comparison retain these qualifications and the population balance proof. Section 5.6 tests a length-proxy version against separate published bacterial cultures, without claiming length is measured mass.

3. Empirical sources and analysis

The applications distinguish a measured resource, a geometric or instrumental proxy, and an inferred composite. Stocks, throughput and accumulated event quantities retain their different physical meanings. Table 1 summarizes the earlier exploratory sources; Section 5 reports frozen predictions, followed by the inverse linguistic application. These convenience cases do not constitute a representative sample of natural systems.

Source	**Analysed material**	**Resource and coordinate**
NOAA GOES XRS, 2022–2024	10,501 events; 1,168 paired primary evaluation events	End fluence; peak irradiance
USGS ComCat, 2010–2024	6,639 selected, rounded moment magnitudes at threshold 5.5	Magnitude-derived energy proxy for both
Hatton et al. (2021)	23 one-decade biomass bins in two related ocean reconstructions	Biomass; body mass
GLOSSAQUA (Ersoy et al., 2025)	1,300 spectrum estimates from 16 study identifiers	Biomass; body mass

Table 1. Earlier exploratory sources. The ocean domains are related, and the aquatic entries are published estimates rather than independent organism observations.

For a size class B_j of logarithmic width w_j, the observed profile is $\widehat{\mathcal{O}}_j = \sum_{i \in B_j} q_i / w_j$. Its normalization to a width-weighted mean of one is an accounting identity, not evidence of flatness. Empty classes, finite-bin integration, missing resources and excluded mass remain explicit. Arithmetic mean cost is the relevant quantity in equation (1); a fitted log cost needs appropriate retransformation.

The exploratory comparisons retain their original domain choices and uncertainty models: calendar blocks for event catalogues, study blocks for pooled aquatic slopes, and stated multiplicative reconstruction errors for ocean biomass. Power-law adequacy uses refitted parametric-bootstrap diagnostics; rounded earthquake magnitudes use their discrete observation grid. Whole-profile equality cannot be inferred from a central slope alone or from pointwise interval overlap. All these methods and their sensitivity results appear in the supporting empirical report.

Input files are checksummed. Local commits document the aquatic and ocean analysis plans and execution order. They do not establish external preregistration or independently verified blinding.

Later studies record when protocols, forecasts and outcomes became available. The linguistic test freezes its vocabulary and model parameters using calibration genres before decoding the evaluation corpus. Its uncertainty summaries remain conditional on those trained models and on the sampled documents.

4. Exploratory allocation results

Application	**Retained result**	**Scientific implication**
Earthquake energy proxy	Gutenberg–Richter $b = 0.998$; year-block 95% interval [0.973, 1.024], versus conversion exponent $\gamma = 1.5$	Disagrees with the specified equal-proxy allocation; energy is inferred from magnitude
Solar flares	Predicted count exponent 1.858 versus observed 2.239; gap 0.382 [0.125, 0.620]	Joint resource, transfer and count-model prediction fails on the selected domain
Upper-ocean reconstruction	Biomass max/min 38.8 across 23 bins; 1.7 across the selected 16-bin plateau	A broad approximate plateau with nonuniformities; re-expression of published evidence
Aquatic spectra	Median normalized slope -1.015; departure interval $[-0.100, 0.010]$ from study-block resampling	Near-neutral central location, without equivalence under the historical criteria

Table 2. Outcomes of the specified exploratory analyses. Intervals retain their original working assumptions; no row supplies a verdict on the general law independent of its resource and observation model.

Several limitations affect interpretation. The solar discrepancy changes with the domain, and its fitted power law fails the stated adequacy test. Ocean uncertainty includes correlated reconstruction errors not captured by the working propagation model; the plateau was selected after inspection. For the full upper-ocean domain, the biomass log-slope is -0.0392 with propagated 90% interval $[-0.058, -0.026]$, outside the stated slope-equivalence tolerance. Aquatic estimates are concentrated in two studies, and 377 records have implausible reported size ranges. The retained sensitivity excludes those records only from range-dependent calculations, without inventing replacement bounds. Reported compatibility fractions are therefore not estimates of

the prevalence of orthopolity in Nature. The supplement preserves every numerical comparison, metadata correction and unresolved observation issue.

5. Frozen-forecast tests

The analyses of Section 4 assess data that had already been inspected. We therefore added tests in which the resource, its cost law, the forecasts and the decision rules are fixed before the evaluated outcomes are decoded, and in which whole units are held out: launches, a calendar year, vessels and selection treatments. Each executed study retains its inputs, frozen protocol, archived algorithms, forecasts and results in an append-only run registry. Three kinds of evidence are kept separate: measurements on engineered systems, an unused observation period of a natural catalogue, and published biological experiments reanalysed retrospectively. Published summaries of those experiments had been read, so their reanalyses are retrospective validations, not blinded tests. A further plant study uses fixed scoring and a development-only geographic split after accidental raw outcome exposure. Its freeze cannot establish that those outcomes were unseen.

5.1 Controlled allocation measurements

In a controlled workload experiment, separately measured memory and CPU costs of matrix tasks predicted completion profiles under assigned quota pairs within four tolerances frozen before 672 validation tasks (workload pilot). A fresh launch exposed a transfer boundary: the original forecast failed its unchanged tolerance in two of four conditions, and local recalibration recovered one (workload transfer). The quotas were acceptance criteria, so these tests concern the transfer of cost models, not allocation.

An allocation test requires the system, not the experimenter, to divide a resource. In a Python runtime whose global interpreter lock serializes execution, worker threads running matrix kernels of five sizes competed for one execution resource (dimensionality intervention). Separate calibration gave CPU cost degrees 1.870 and 2.686 for the two kernels. The calibrated mean-cost curves predicted complete CPU and job profiles before 16 allocation trials, with maximum CPU-share errors of 0.0035–0.0063 and job-share total-variation errors of 0.0021–0.0124. Replacing the largest-class thread with a second smallest-class thread produced the predicted CPU-share change within 0.0078. A unit-cost alternative had count errors of 0.126–0.273. This success concerns one engineered allocation mechanism; it identifies no geometric dimension and no tendency of natural systems.

5.2 Calibrated decisions and an unused observation period

Complete-profile equivalence decisions were calibrated on 180,000 independently generated datasets in 90 conditions (profile calibration). Simultaneous bands met the declared gate for dense, bounded, independent blocks at 48 blocks, with minimum coverage 93.95%, but failed at 12 and 24 blocks. With twelve serially dependent blocks, wrong-departure rates reached 43.75%. These failures define observation designs for which no neutrality verdict is available, whatever a nominal interval suggests.

For solar flares, measured rise-phase fluence costs, six logarithmic classes, three forecasts, a tolerance and an application gate were committed before the 2025 NOAA records were acquired (solar validation). Of 3,277 events, 356 met the domain rules. Logarithmic neutrality predicted the class counts much better than linear neutrality, with count total variation 0.125 against 0.573; historical counts did better still, at 0.022. The point profile departed from the $\ln 1.5$ margin, but

the calibrated decision remained unresolved, as declared before acquisition, and the instrument mix changed between periods.

5.3 Published biological measurements

The *Synechococcus* comparison demonstrates that a size-based cost can transfer for one resource and fail for another in the same organisms: cubic diameter predicted held-out 25 °C carbon quotas within a typical factor of 1.14, while strain means predicted N and P better (cost transfer).

The positive biological allocation result concerns separately grown populations. Thirty size-selected *Dunaliella* lineages were regrown in one shared F/2 medium. Across a 10.4-fold range of mean cell volume, replete capacity in total biovolume was constant to within 3%, and cell number scaled as $V^{-1.02}$ (Figure 4). Each held-out selection treatment was predicted from the other two. The frozen equal-biovolume law gave held-out errors of 0.124 and 0.179 log units, outperforming a fitted size law and equal cell number. This establishes conditional budget closure across cultures; it does not demonstrate exchange among coexisting size classes. Regrowth after P deprivation retained a size-dependent overshoot (size budget).

The chemostat response supplies a contrasting dynamical test. A nitrogen pulse changed algal resource composition, but no frozen prediction improved appreciably on persistence in twelve held-out polyculture vessels. The two-budget C:N reweighting rule predicted too little redistribution, 0.05–0.12 versus observed departures of 0.29–0.49, and the predicted direction held in only 4 of 12 vessels. No vessel recovered its pre-pulse composition within twelve days (chemostat response). This particular constraint model failed; its frozen result is retained.

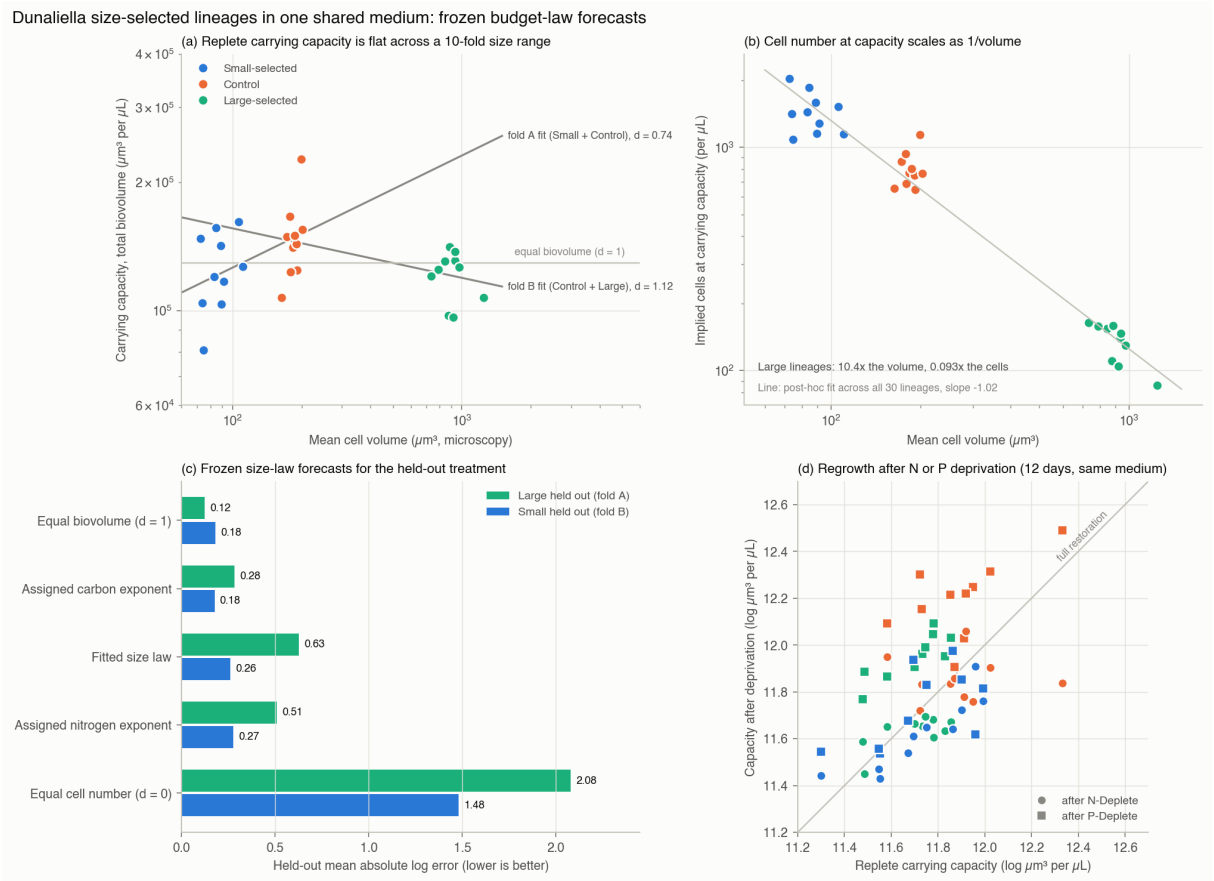


Figure 4. Size-selected Dunaliella lineages regrown in one shared medium, from the data of Malerba et al. (2018). (a) Replete carrying capacity in total biovolume against mean cell volume, with the two cross-fitted size laws and the equal-biovolume law. (b) Implied cell number at capacity. (c) Held-out errors of the frozen forecasts. (d) Capacity after N or P deprivation against replete capacity. Generated from retained outputs by report_dunaliella_size_budget.py.

5.4 Directly weighed coexisting plant stocks

Ten harvested herbaceous plots provide direct aboveground dry masses for coexisting ramets or stem clusters (Dillon et al., 2019; plant study). Forest and desert allometric masses are excluded. Five Ohio plots supply development fits; five Colorado plots supply evaluation. The protocol, algorithms and forecasts were committed before formal scoring, but raw rows had already been exposed during source inspection. This is a retrospective transfer comparison. It identifies neither a limiting nutrient nor an opportunity budget; $q(m) = m$ defines the measured stock rather than testing an independent cost law.

The frozen domain is 0.01–100 g in eight half-decade bins. Thirty missing masses, 144 subthreshold ramets and one above-domain ramet remain in the membership ledger. The above-domain ramet contains 24.50% of one plot's known biomass. Results describe the declared domain.

Mean biomass total variation is 0.470 for the transferred bounded Pareto template and 0.493 for logarithmic neutrality (Figure 5). No template is best in every plot. Trained count models score 0.260–0.271, compared with 0.495 for logarithmic neutrality. These are point comparisons; no calibrated ecological superiority or equivalence verdict follows.

Synthetic censuses explain an observation limitation. Under exactly equal expected biomass, 160 independent ramets produce mean realized stock TV 0.441 and an empty bin in every simulated census. The largest bin's average normalized share is 3.13%, while its normalized expected stock is 12.50%: $E[R_i/\sum_j R_j] \neq E[R_i]/\sum_j E[R_j]$. Repeating masses in blocks of twenty also makes an iid 95% envelope fail severely. These simulations diagnose finite-census normalization and dependence sensitivity, without identifying the ecological dependence model. Expected neutrality remains unresolved. The supporting empirical report retains all scores, coverage calculations and sensitivities.

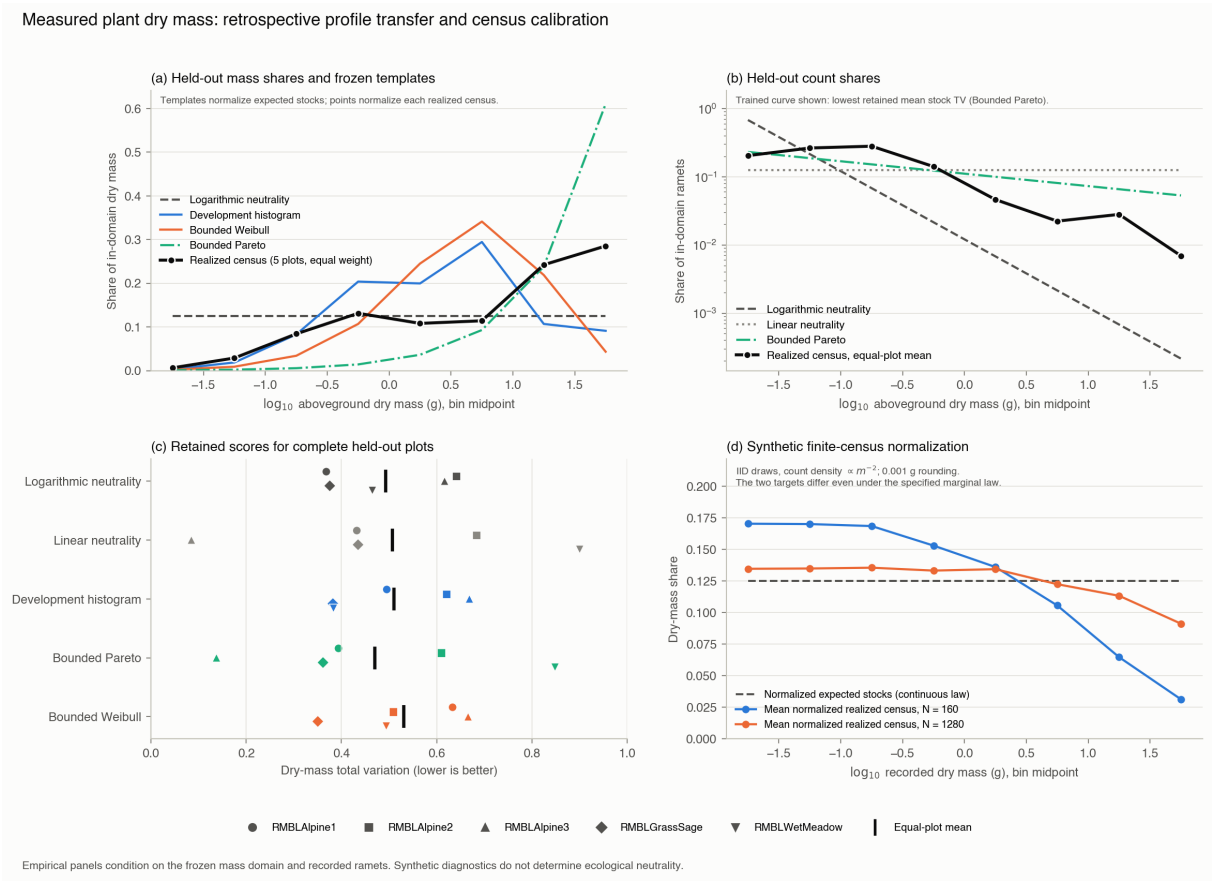


Figure 5. Directly weighed plant profiles and synthetic observation limits. (a) Equal-plot mean Colorado biomass shares and transferred stock templates. (b) Count shares and retained count forecasts. (c) Each plot's biomass discrepancy and the five means. (d) Mean normalized census shares under iid logarithmic neutrality, compared with normalized expected stocks. All panels read retained outputs; report_plant_biomass_profile.py performs presentation only. Prior raw exposure and excluded biomass are retained.

5.5 Inferring a linguistic resource and predicting withheld genres

The linguistic study turns the inverse proposal into a measured transfer comparison (protocol and report). Word occurrences come from the pinned Universal Dependencies English EWT corpus, release 2.18; letter counts L come from original sentence text and phoneme counts P from an independently pinned CMU pronunciation dictionary. The model uses a counting measure on word types, with no fitted rank transformation or post-hoc constraint factor.

Training uses only email, weblog and newsgroup texts: 83 documents and 83,635 retained tokens determine a vocabulary of 2,623 types with at least five training occurrences and a dictionary pronunciation. All six forecasts share this vocabulary. Conditional multinomial likelihood with nonnegative exponents gives

$$\widehat{Q}_w \propto L_w^{0.5153} P_w^{1.8275}, \qquad \widehat{p}(w) \propto \widehat{Q}_w^{-1}.$$

Both exponents are interior. The log-feature correlation is 0.871 and the model information condition number is 11.65; these diagnostics describe the fitted model, without treating dependent tokens as independent measurements of parameter precision. The protocol, code and fitted forecasts were hashed before the official test file was acquired or decoded.

The primary target comprises withheld review and answer genres: 253 documents, 7,329 retained tokens and 77.65% vocabulary coverage after exact-duplicate exclusion. The secondary target contains 63 new documents from the training genres, with 9,618 retained tokens and 79.35% coverage. Ninety-three sentences matching original training text were excluded by the frozen rule, and no training/test document IDs overlap. Coverage is conditional on the tokenizer and fixed vocabulary; excluded tokens remain in the ledger.

Frozen forecast	**New genres**	**Represented genres**
Uniform word types	7.872	7.872
Inverse letter count	7.405	7.457
Fitted letter power	7.239	7.328
Fitted letter/phoneme product	7.055	7.189
Exponential letter cost	7.195	7.304
Smoothed training frequencies	6.136	6.199

Table 3. Held-out conditional log loss in nats per retained token; lower is better. Every parameter and the vocabulary are fixed from training.

The composite improves on letter power by 0.1843 nats per token in new genres; the paired document-bootstrap 95% interval for composite minus letter-power loss is $[-0.1967, -0.1696]$. The secondary improvement is 0.1387, with interval $[-0.1512, -0.1251]$, and the point difference favors the composite in all five genres. The 2,000-resample intervals condition on the fitted forecasts and exchangeability of documents within each evaluation group. They omit training uncertainty and dependence between authors, threads or websites.

The gain demonstrates transferable information in the combined symbolic features (Figure 6). Training word frequencies nevertheless predict much better. Only 83 distinct (L, P) pairs occur among the 2,623 vocabulary types, so the resource model must give many different words identical probabilities. Observed primary letter-resource and inferred-cost shares remain far from uniform over word types, with descriptive total variations 0.693 and 0.689. Within every individual genre the composite distance is larger than the letter-resource distance, despite the small decrease after pooling. These finite-corpus discrepancies are not calibrated neutrality tests. The fitted combination is a predictive candidate; neither equal resource allocation nor a unique physical

linguistic cost has been established. Sonority, duration, context and semantic requirements remain unmeasured here.

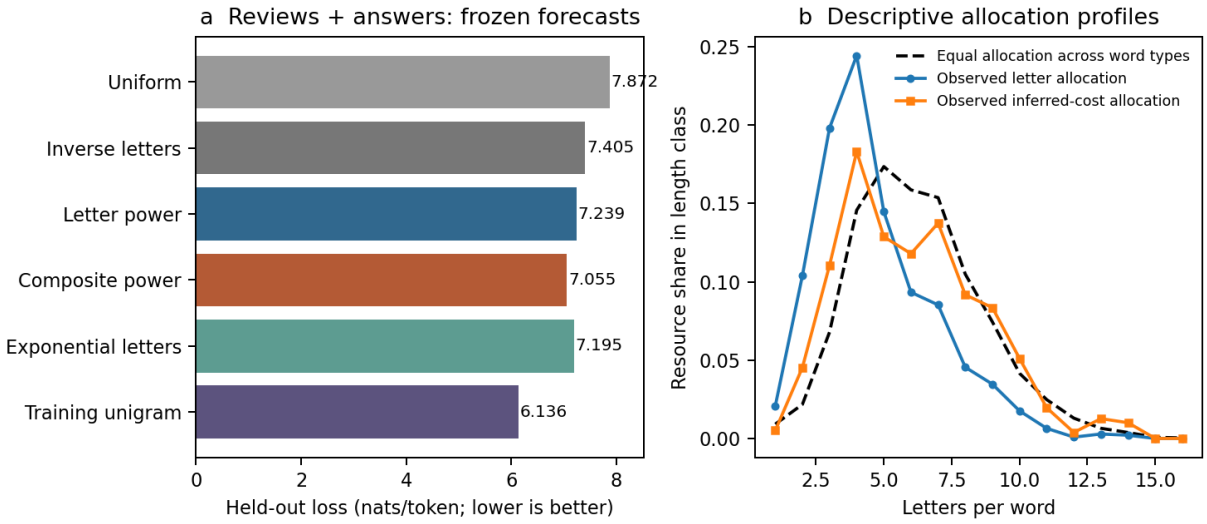


Figure 6. Frozen linguistic predictions. (a) Loss on withheld review and answer genres, including all six declared alternatives. (b) Observed letter and inferred-cost allocation aggregated by word length. Equal allocation over word types gives the dashed type-multiplicity profile, not a flat profile over length. Aggregation conceals variation among words of the same length. The fitted composite improves prediction over letter-only costs but trails training word frequencies; its resource profile remains descriptive.

5.6 Birth-calibrated predictions of bacterial population summaries

Gangan and Athale (2017) provide mother-machine birth/division lengths and separate mid-log batch snapshots of *E. coli* MG1655 in LB and M9+succinate. We use their printed fitted lognormal birth moments to predict batch mean length and coefficient of variation (CV) from Section 2.12. For birth mean m and squared CV c^2,

$$E_P[x] = \frac{2m \ln 2}{1 + c^2}, \qquad \mathrm{CV}_P^2 = \frac{1 + c^2}{2(\ln 2)^2} - 1.$$

Calibration, algorithms and all forecasts were frozen before acquiring the numerical target figure. Published qualitative results had already been read: this is a retrospective summary transfer, not external preregistration or global blinding. No target scale or shape is fitted. The raw workbook could not be downloaded; official source images, transcriptions and receipts are retained in the study report.

Endpoint	Published	Population	Lineage
LB mean	3.2842	3.3737	3.9517
LB CV	0.3957	0.4139	0.4126
M9 mean	2.0094	2.0184	2.1515
M9 CV	0.2146	0.2568	0.2549

Table 4. Frozen predictions against published fitted summaries. Means are in micrometres; CV is dimensionless. M9 denotes M9+succinate.

Mean errors are 2.72% and 0.447%, versus 20.32% and 7.07% for the lineage comparator. CV is overpredicted by 4.61% and 19.69%; the lineage comparator is slightly closer on CV. The

declared requirement of improvement on both endpoints therefore fails in both media. This is not statistical model rejection; the two CV formulas are structurally very close. All four forecasts, including fixed-birth and division-calibrated sensitivities, remain in the report; division calibration improves LB but worsens the M9 mean. These are comparisons to rounded fitted summaries, with no sampling confidence interval or full-distribution test.

The calibration also diagnoses model limitations: division means are 0.9644 and 1.0849 times twice the birth means, whereas exact stationary symmetric division requires equality. Common individual growth and comparable old-pole physiology are not established. Length is only a resource proxy: even at constant diameter d, spherocylinder volume is proportional to $L - d/3$. The mean transfer is a quantitative success of the conditional approximation; the variability and calibration mismatches prevent treating it as a validated biomass-allocation profile. Resolving these departures requires additional independent information about the dynamics and observation process.

5.7 Geometric resource predictions and seismic eligibility

Separate repository registrations compare physical resource readings using earthquake aftershocks, flare ribbons and lake geometry. They concern log-orthopolic allocation per logarithmic size coordinate. Inputs, estimator amendments, prior exposure and decision rules are retained in the test ledger. These studies supply negative or qualified comparisons; the literal registered decisions remain distinct from whether an interval excludes an individual predicted value.

Comparison	**Result**	**Scope of the decision**
Aftershock productivity: rupture area predicts $\alpha = 1$, moment predicts 1.5	Primary $\alpha = 0.63$, 95% interval $[0.51, 0.71]$; rupture-length-window sensitivity 0.90, $[0.76, 0.97]$	Both predictions excluded under these counting schemes; registered verdict inconclusive; strong window dependence
Flare ribbon area: density exponents 2 and 7/3	$\alpha = 3.163$, $[2.962, 3.457]$	Both excluded; registered verdict inconclusive
Global lake radius: survival exponent $\zeta_R = 2$	Original geometry 1.930, iid $[1.919, 1.941]$; local-geometry follow-up 1.859, basin $[1.785, 1.924]$	Original iid equivalence disappears with physical local geometry; follow-up inconclusive against the $[1.9, 2.1]$ margin

Table 5. Retained geometric predictions. Intervals condition on each study's declared sampling and measurement assumptions; they do not establish a universal resource law.

The lake follow-up independently changes projection and quantifies basin dependence. Two local projections agree closely, identifying the original global projection as a material source of the change. Exact 2 lies outside the basin interval, while practical equivalence and practical departure are both unestablished under the frozen margin. Invalid source topology, unmatched basin outlets and an iid goodness-of-fit diagnostic remain qualifications. The separately frozen UK result likewise does not establish a robust spatial rejection: its iid interval excludes 2, but its spatial interval includes it.

The seismic comparison predicts $b = D_2/2$ for rupture territory, using hypocentre correlation dimension, against $b = 1$ independent of that dimension. The Southern California qualification uses published sequence memberships. Forty sequences have at least 200 raw events; quality, completeness and location-resolution gates reduce this to six, below the registered minimum ten. No b or D_2 was fitted.

A subsequent CLASS/HORUS source audit establishes 295,193 unambiguous Italian location/provenance links, of which only 699 have direct moment-tensor magnitudes. The calibrated

proxies require an observation model. A separately frozen synthetic check, using the published conversions of Lolli et al. (2020), fails its necessary validation gate in 45 of 56 required scenarios, including 30 of 32 noiseless proxy cases. For example, true $b = 1.5$ gives mean 1.158 under one earlier-ML conversion and lower-support choice; exact source-bin integration gives 1.158 as well. Conversion leaves gaps in rounded magnitude classes which the uniform-bin correction does not represent. This is a controlled counterexample to the estimator's adequacy, not an observed Italian exponent or a test of either physical relation. The Italian attempt therefore stops before real sequence construction or response fitting. A revised likelihood would need to represent original magnitude bins, detection and calibration uncertainty before another physical comparison.

5.8 What the tests establish

The positive results concern conditional budget closure: a separately measured or geometric cost, declared shared resource conditions, and abundance or resource shares predicted from them. The runtime experiment and the algal lineages are the clearest cases. In the lineages, volume scaling supplies the cost exponent; the empirical content is the approximately constant attained biovolume across separately grown lineages. The capacity theorem makes inverse-volume abundance conditional on that common attained budget. The biological observation concerns whether the lineages actually reach it. A rule that goes beyond closure, by predicting how a restriction redistributes resource among coexisting classes, failed in the grazed, dynamically changing food webs tested. The plant study now observes coexisting class stocks directly, but it does not establish equal expected allocation: the flat template is not best for biomass or counts, and the ecological observation law is uncalibrated. No natural size-class study in Section 5 has independently identified neutral eligibility and a design sufficient for the expected-allocation verdict. Identifying such a regime, or independently characterizing its constrained allocation, remains a central empirical task. The linguistic study separately shows a gain from an inferred composite in a frozen genre transfer. Its lexical baseline and uneven resource shares retain the limits of that candidate. The bacterial comparison adds a mechanism-derived prediction between separate experimental settings: mean length transfers closely, while variability and the joint comparison fail. Its resource proxy and unverified growth assumptions leave the natural-allocation qualification task open.

6. Discussion

Orthopolity has an exact mathematical core in finite capacity and complete resource hierarchies. The hierarchy theorem establishes inverse mean-resource incidence across full descriptions of the same resource, including nested units. Its coverage factor quantifies incomplete and repeated accounting. This distinguishes a result guaranteed by the construction from a physical claim that comparable natural concentrations receive comparable resources. Both belong in the framework: the exact result anchors the generalization and specifies which premise a natural application must establish.

The general natural-law formulation concerns resource allocation. Its neutral expression is equal effective resource per physically specified comparison measure; its observed expression includes the effects of the system and medium. Effective resources can combine several constituents nonlinearly. A distribution that is not flat in one separately measured constituent can therefore reflect a composite allocation, a constraint, or both. These possibilities become explanations when their combination rules and consequences are specified. The law does not identify every visibly unequal distribution with failure of the underlying tendency.

The class-exchange construction provides an explicit model in which an unequal stationary profile contains active opposing contributions. Removing the constraint predicts recovery without resetting the state. The electromagnetic examples establish related physical realizations with different observables: nested throughput, disjoint shell energy, and thermal energy per mode. Their measures and equality conditions follow from their respective equations. Quantization restricts available mode energies; it is a different kind of constraint from an additive transport bias. The framework can encompass these mechanisms without asserting that their dynamics are identical. Growth and division add an observation-sensitive realization: mass weighting of population counts gives a chronological lineage law under the stated conditions. Birth/division events then determine the allocation envelope. The bacterial transfer supplies a numerical mean prediction and a variability discrepancy that an unconstrained inverse-square label would conceal.

The inverse direction is central to the proposed law's scientific use. An abundance profile constrains effective resource divided by its allocation distortion, conditional on the comparison measure. A Pareto exponent can identify its scaling even when the constituent resource combination is unknown. It does not uniquely determine that combination, but shared constitutive families and independently characterized environments can make the inverse problem identifiable. The result in Section 2.3 specifies when composition or a target prediction is identified, and its left-null contrasts make a shared resource model testable across environments. Correlated errors and nearly proportional constituent scalings determine how precise that test can be. City-size scaling motivates applying this strategy to coupled urban requirements; no city resource combination has yet been measured or fitted here.

The linguistic comparison executes a complementary inverse application using direct symbolic counts. Its letter/phoneme combination improves frozen predictions in withheld genres, with the same signs of improvement in represented genres. The stronger lexical baseline and the large observed allocation discrepancies also show that these two length features do not explain complete word usage. These findings motivate independently characterized contextual, temporal or phonetic requirements. Adding an arbitrary fitted constraint after seeing a residual would not identify such a requirement.

The empirical studies contribute different kinds of evidence. Separately calibrated costs and shared resource conditions support budget-closure predictions in the engineered allocation system and the size-selected algal lineages. The chemostat cost-ratio rule fails its declared redistribution predictions, and plant stock/count comparisons remain mixed. The FIRAS transformation preserves the published quantum-spectrum result and its reconstruction assumptions. The natural size-class studies do not yet identify a neutral regime and its effective resource with sufficient observation information for a general expected-allocation verdict. A nonlinear composite hypothesis opens further research; it does not retrospectively change these studies' frozen predictions or outcomes.

The proposal builds on established work. Reversible transport and equipartition supply mechanisms and limiting cases; ecological size-spectrum models already connect resource use and abundance (Cuesta, Delius and Law, 2018; Arranz et al., 2022). Independent metabolic measurements already support cost-to-capacity predictions (Marshall et al., 2022). Proportional-growth models address city-size scaling (Gabaix, 1999). The lineage/population transformation and inverse-square cell-cycle limit likewise have established derivations (Genthon, 2022). The contribution sought from orthopolity is a general allocation account that connects these results, organizes their constraints, and generates further forward and inverse predictions. These relationships require explicit derivations and comparisons, not an assumption that naming a common pattern establishes a common microscopic mechanism.

The inverse formulation also has established statistical foundations. With deterministic constituent requirements, a product resource gives

$$p_\theta(k \mid s) = Z_s(\theta)^{-1} \exp\left(\ln a_s(k) - \sum_\ell \theta_\ell \ln X_{s\ell}(k) \right)$$

relative to the declared measure. This is an exponential family. Its likelihood, rank conditions and error propagation use standard methods; the physical content lies in the specified resources, constraints and shared composition across conditions. Inverse optimization and inverse statistical mechanics already infer governing quantities from outcomes (Keshavarz, Wang and Boyd, 2011; Habeck, 2014). The left-null relations in Section 2.3 state what the proposed composition must predict across environments. A fitted resource that reproduces its calibration profile has not yet passed that test.

The next empirical advance should connect a resource combination to independently measured constraints or physical requirements across conditions, extending beyond the linguistic symbolic-count pilot. Specify the object partition, comparison measure, constituent requirements and budgets, and observable target; use a calibration subset to infer the combination and its uncertainty; then predict another distribution or a response to a specified change. Variation in resource requirements must distinguish the candidate combinations, and the observation model must account for dependence and finite samples. A deeper microscopic mechanism can strengthen the explanation, but independent predictive success can also support an allocation law. This programme tests and extends the scope of the proposed law while using distributions to learn about the resources through which natural systems are organized.

7. Conclusion

This paper establishes the mathematical foundation of orthopolity in finite resource capacity and complete resource hierarchies. Larger requirements impose inverse capacity bounds; complete descriptions of the same resource obey exact inverse relations between count and mean resource per unit. The result includes nested units and quantifies the effects of incomplete coverage and overlap. Requiring a composite resource to remain unchanged under subdivision further restricts its admissible form.

The general natural-law formulation states that Nature tends to distribute resources equally among concentrations of those resources. Its expression depends on the system's construction, medium, interactions and comparison measure. Nonlinear effective resources remain possible within their declared physical partitions and constitutive conditions. Forward prediction and inverse inference connect that formulation to observable systems.

The neutral allocation relation gives inverse-cost abundance under a declared measure. The class-exchange model supplies a mechanism, a constrained unequal equilibrium with opposing contributions, and quantitative recovery. Electromagnetic transport and thermal radiation supply worked physical realizations and explained departures. The retained observational studies contribute conditional successes, failed predictions and unresolved questions with their original scope preserved. A Pareto profile supplies a conditional constraint on effective-resource scaling; additional environments can identify and test candidate nonlinear combinations. The linguistic transfer provides an executed inverse application: a combined symbolic cost improves prediction over letter-only models, while a lexical baseline and residual allocation differences delimit its explanatory scope. The bacterial comparison connects known growth-fragmentation dynamics to a resource measure and independently calibrated population means; overpredicted variability retains the approximation's limits.

The exact geometric results and specified physical mechanisms establish the stated capacity, hierarchy and dynamical relations. The broader natural-law claim concerns their reach across independently characterized systems. Its scientific significance rests on that explanatory reach

and on further consequences that can be tested. Identifying resources, coverage and constraints independently connects the simple core to those consequences.

Data and code availability

Code, frozen inputs, configurations, and generated results are in this repository. The make all target verifies input checksums, runs tests, and regenerates the empirical analyses; see the README for setup. Pilot estimates originate in results.json; distribution diagnostics in gof.json; aquatic median, compatibility fractions, and ocean intervals in independent.json. Remaining JSON files contain explicitly exploratory diagnostics. The studies of Section 5 are registered in an append-only run registry with their inputs, frozen protocols, archived algorithms and outputs. The class-exchange demonstration and retrospective FIRAS calculation are also registered; their reports give offline reproduction commands and interpretation limits. The inverse-resource simulation and linguistic transfer likewise retain configurations, algorithms, predictions, all scored outcomes and source hashes. The linguistic protocol and training fit were frozen before test-file acquisition; this is a retrospective public-corpus study, not external preregistration. The bacterial summary transfer retains official figure images, manual numerical transcriptions, calibration diagnostics and four frozen forecasts. Its numerical target figure was retrieved after the freeze; qualitative outcomes were already known. The geometric tests, seismic eligibility assessment and Italian synthetic magnitude validation retain separate configurations, source receipts and reproduction targets. Their outputs preserve failures and stopping decisions; no real Italian b/D2 result is implied by the simulation record. Plant sources are pinned to author commit defccc3dcbbbf3ba57ff1572377de88fba83ff7f, with acquisition checksums and complete inclusion ledgers. The registry-verify target audits the registry offline, and each study report gives its replay command; source errata found during those audits are recorded there. Source-specific conditions are recorded in data/NOTICE.md; there is no blanket licence for all inputs. The supplied 2017 manuscript was inspected privately and is not redistributed.

References

Arranz, I., Fournier, B., Lester, N. P., Shuter, B. J., and Peres-Neto, P. R. (2022). Species compositions mediate biomass conservation: The case of lake fish communities. *Ecology*, 103, e3608. https://doi.org/10.1002/ecy.3608.

Bishop, C. J., and Peres, Y. (2017). *Fractals in Probability and Analysis.* Cambridge University Press. Section 1.1. https://www.math.stonybrook.edu/~bishop/fractalbook.pdf.

Boyd, S., and Vandenberghe, L. (2004). *Convex Optimization.* Cambridge University Press. Sections 4.7 and 5.6. https://web.stanford.edu/~boyd/cvxbook/.

CMU Sphinx contributors (accessed 7 October 2026). *CMU Pronouncing Dictionary.* Pinned source.

Clauset, A., Shalizi, C. R., and Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51, 661–703. https://doi.org/10.1137/070710111.

Condon, J. J., and Ransom, S. M. (2016). *Essential Radio Astronomy.* Princeton University Press. Chapter 2, radiation and absorption. https://www.cv.nrao.edu/~sransom/web/Ch2.html.

Cuesta, J. A., Delius, G. W., and Law, R. (2018). Sheldon spectrum and the plankton paradox: two sides of the same coin—a trait-based plankton size-spectrum model. *Journal of Mathematical Biology*, 76, 67–96. https://arxiv.org/abs/1607.04158.

Damuth, J. (1981). Population density and body size in mammals. *Nature*, 290, 699–700. https://doi.org/10.1038/290699a0.

Dillon, K. T., et al. (2019). On the relationships between size and abundance in plants: beyond forest communities. *Ecosphere*, 10, e02856. https://doi.org/10.1002/ecs2.2856.

Edwards, A. M., Robinson, J. P. W., Plank, M. J., Baum, J. K., and Blanchard, J. L. (2017). Testing and recommending methods for fitting size spectra to data. *Methods in Ecology and Evolution*, 8, 57–67. https://doi.org/10.1111/2041-210X.12641.

Edwards, A. M., Robinson, J. P. W., Blanchard, J. L., Baum, J. K., and Plank, M. J. (2020). Accounting for the bin structure of data removes bias when fitting size spectra. *Marine Ecology Progress Series*, 636, 19–33. https://doi.org/10.3354/meps13230.

Eeckhout, J. (2004). Gibrat's law for (all) cities. *American Economic Review*, 94(5), 1429–1451. https://doi.org/10.1257/0002828043052303.

Ersoy, Z., et al. (2025). GLOSSAQUA: A global dataset of size spectra across aquatic ecosystems. *Ecology*, 106, e70050. https://doi.org/10.1002/ecy.70050.

Fabbri, R., and Oliveira Jr., O. N. (2017). *A simple model that explains why inequality is ubiquitous.* Supplied manuscript dated 17 March 2017, 13 pp.; publication status unverified.

Fabbri, R. (2024). The Orthopolity cosmological principle and the Natural distribution law. Author's essay, 14 August. https://ttm.github.io/2024/08/14/power.html.

Fixsen, D. J., Cheng, E. S., Gales, J. M., Mather, J. C., Shafer, R. A., and Wright, E. L. (1996). The cosmic microwave background spectrum from the full COBE FIRAS data set. *The Astrophysical Journal*, 473, 576. https://doi.org/10.1086/178173. Source article: https://arxiv.org/abs/astro-ph/9605054.

Gabaix, X. (1999). Zipf's law for cities: An explanation. *The Quarterly Journal of Economics*, 114(3), 739–767. https://doi.org/10.1162/003355399556133.

Gaedke, U. (1993). Ecosystem analysis based on biomass size distributions: A case study of a plankton community in a large lake. *Limnology and Oceanography*, 38, 112–127. https://doi.org/10.4319/lo.1993.38.1.0112.

Gangan, M. S., and Athale, C. A. (2017). Threshold effect of growth rate on population variability of *Escherichia coli* cell lengths. *Royal Society Open Science*, 4, 160417. https://doi.org/10.1098/rsos.160417.

Genthon, A. (2022). Analytical cell size distribution: lineage-population bias and parameter inference. *Journal of the Royal Society Interface*, 19, 20220405. https://doi.org/10.1098/rsif.2022.0405. Revised exposition: https://arxiv.org/abs/2206.06146v2.

Harcourt, R., Garcia, N. S., and Martiny, A. C. (2024). *Synechococcus* batch culture data (cell quotas and ratios (C, N, P), size, and diameter) from laboratory experiments in 2021 to 2022 with related isolates cultured across a range of temperatures. BCO-DMO dataset 926311, version 1. https://doi.org/10.26008/1912/bco-dmo.926311.1.

Hatton, I. A., Heneghan, R. F., Bar-On, Y. M., and Galbraith, E. D. (2021). The global ocean size spectrum from bacteria to whales. *Science Advances*, 7, eabh3732. https://doi.org/10.1126/sciadv.abh3732.

Habeck, M. (2014). Bayesian approach to inverse statistical mechanics. *Physical Review E*, 89, 052113. https://doi.org/10.1103/PhysRevE.89.052113.

Haus, H. A., and Melcher, J. R. (1989). *Electromagnetic Fields and Energy.* Prentice-Hall. Section 11.2, Poynting's theorem. https://web.mit.edu/6.013_book/www/chapter11/11.2.html.

Ioannides, Y. M., and Skouras, S. (2013). US city size distribution: Robustly Pareto, but only in the tail. *Journal of Urban Economics*, 73(1), 18–29. https://doi.org/10.1016/j.jue.2012.06.005.

Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106, 620–630. https://doi.org/10.1103/PhysRev.106.620.

Jaynes, E. T. (1968). Prior probabilities. *IEEE Transactions on Systems Science and Cybernetics*, 4(3), 227–241. https://bayes.wustl.edu/etj/articles/prior.pdf.

Keshavarz, A., Wang, Y., and Boyd, S. (2011). Imputing a Convex Objective Function. *Proceedings IEEE Multi-Conference on Systems and Control*, 613–619. https://stanford.edu/~boyd/papers/imputed_objective.html.

Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*, second edition, with contributions by E. L. Wilmer. American Mathematical Society. https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf.

Lolli, B., Randazzo, D., Vannucci, G., and Gasperini, P. (2020). The Homogenized Instrumental Seismic Catalog (HORUS) of Italy from 1960 to Present. *Seismological Research Letters*. https://doi.org/10.1785/0220200148.

Malerba, M. E., Palacios, M. M., and Marshall, D. J. (2018). Do larger individuals cope with resource fluctuations better? An artificial selection approach. *Proceedings of the Royal Society B*, 285, 20181347. https://doi.org/10.1098/rspb.2018.1347. Data: https://doi.org/10.5061/dryad.4mh47r7.

Marshall, D. J., et al. (2022). Long-term experimental evolution decouples size and production costs in *Escherichia coli*. *Proceedings of the National Academy of Sciences*, 119, e2200713119. https://doi.org/10.1073/pnas.2200713119.

Newman, M. E. J. (2005). Power laws, Pareto distributions and Zipf's law. *Contemporary Physics*, 46, 323–351. https://arxiv.org/abs/cond-mat/0412004.

Piantadosi, S. T., Tily, H., and Gibson, E. (2011). Word lengths are optimized for efficient communication. *Proceedings of the National Academy of Sciences*, 108(9), 3526–3529. https://doi.org/10.1073/pnas.1012551108.

NASA LAMBDA (accessed 6 October 2026). COBE FIRAS CMB monopole spectrum, version 1. Product description and reconstruction provenance. https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_spect.html.

Planck, M. (1901). Ueber das Gesetz der Energieverteilung im Normalspectrum. *Annalen der Physik*, 309, 553–563. https://doi.org/10.1002/andp.19013090310.

Perkins, D. M., et al. (2018). Bending the rules: exploitation of allochthonous resources by a top-predator modifies size-abundance scaling in stream food webs. *Ecology Letters*, 21, 1771–1780. https://doi.org/10.1111/ele.13147.

Poynting, J. H. (1884). On the transfer of energy in the electromagnetic field. *Philosophical Transactions of the Royal Society of London*, 175, 343–361. https://doi.org/10.1098/rstl.1884.0016.

Sheldon, R. W., Prakash, A., and Sutcliffe, W. H., Jr. (1972). The size distribution of particles in the ocean. *Limnology and Oceanography*, 17, 327–340. https://doi.org/10.4319/lo.1972.17.3.0327.

Tao, T. (2011). *An Introduction to Measure Theory*. Graduate Studies in Mathematics, 126. American Mathematical Society. Exercise 1.4.34. https://terrytao.wordpress.com/wp-content/uploads/2012/12/gsm-126-tao5-measure-book.pdf.

Tong, D. (n.d.). *Statistical Physics*. University of Cambridge Part II Mathematical Tripos lecture notes, Sections 1.3, 2.2.1 and 3.2. Accessed 6 October 2026. https://davidtong.org/pdfs/teaching/statistical-physics/statphys.pdf.

Visser, M. (2013). Zipf's law, power laws and maximum entropy. *New Journal of Physics*, 15, 043021. https://arxiv.org/abs/1212.5567.

Wojcik, L. A. M., Pfennig, A., Flamm, S., Klauschies, T., Rosenbaum, B., Weithoff, G., and Gaedke, U. (2025). Top-down control and species composition influence nonlinearly the short-term response of experimental food webs to a nutrient pulse perturbation. *Proceedings of the Royal Society B*. https://doi.org/10.1098/rspb.2025.1969. Data: https://doi.org/10.5061/dryad.51c59zwj5.

Universal Dependencies contributors (2026). *English EWT*, release 2.18. Corpus description and pinned source.